GENERAL SAFETY AND SECURITY HOME SYSTEM

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ABSTRACT

This project addresses the growing need for comprehensive home and personal safety solutions in response to common but critical hazards such as oxygen depletion in enclosed spaces and unauthorized home access. Many households face risks from gas heaters used during sleep or poor ventilation during showers, which can lead to dangerous drops in oxygen levels. Additionally, homeowners often lack efficient systems to monitor or control access to their property, especially during prolonged absences.

To solve these issues, the project proposes a smart, integrated alert and recognition system that combines environmental sensing, facial recognition, and smart control through a mobile application. The system detects unsafe oxygen levels in real time and sends immediate alerts, restricts access using face recognition technology, and simulates occupancy to deter potential intrusions. All components are managed via an easy-to-use application that allows users to monitor and customize the system, providing a unified solution for enhancing personal and home safety.

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Finally, we extend our heartfelt appreciation to our families and friends for their constant support, understanding, and encouragement, which sustained us through the challenges of this project.

DEDICATION

To our families, whose unwavering love, sacrifice, and steadfast support have been the cornerstone of our journey. Your faith in us has provided strength and purpose through every challenge we encountered.

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ABBREVIATIONS

Table 0.1: Abbreviation Table.

Abbreviation	Full Form
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
API	Application Programming Interface
API Key	Application Programming Interface Key
ASCII	American Standard Code for Information Interchange
BGR	Blue-Green-Red
CLI	Command-Line Interface
CNN	Convolutional Neural Network
Colab	Google Colaboratory
COM	Common Terminal
CO_2	Carbon Dioxide
CPU	Central Processing Unit
CV2	OpenCV (Open-Source Computer Vision Library)
DAC	Digital-to-Analog Converter
DLIB	Deep Learning Library
ESP32	Espressif Systems 32-bit Microcontroller
FCM	Firebase Cloud Messaging
FPS	Frames Per Second
GND	Ground
GPIO	General Purpose Input/Output
HOG	Histogram of Oriented Gradients
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment

IoT	Internet of Things
JD	Jordanian Dinar
JPEG	Joint Photographic Experts Group
JSON	JavaScript Object Notation
LED	Light Emitting Diode
ML	Machine Learning
MQ-2	Metal Oxide Semiconductor Sensor Type 2
MQTT	Message Queuing Telemetry Transport
NC	Normally Closed
NO	Normally Open
O_2	Oxygen
OAuth2	Open Authorization 2.0
OS	Operating System
PIR	Passive Infrared (sensor)
PNG	Portable Network Graphics
PWM	Pulse Width Modulation
QoS	Quality of Service
RAM	Random Access Memory
ResNet	Residual Neural Network
REST	Representational State Transfer
RESTful	Representational State Transfer
RGB	Red-Green-Blue
SDK	Software Development Kit
SHA-256	Secure Hash Algorithm 256-bit
SVM	Support Vector Machine
UI	User Interface

URL	Uniform Resource Locator
USB	Universal Serial Bus
VCC	Voltage Common Collector
VR	Variable Resistor (Potentiometer)
VS Code	Visual Studio Code
WROOM	Wireless Room Module (ESP32 variant)

Chapter 1: Introduction

Overview

In light of the growing reliance on technology to provide smart solutions for daily life, this project emerges as an integrated security system aimed at protecting individuals and their homes from various environmental and security threats. The system addresses recurring but often overlooked issues, such as the risk of oxygen depletion in enclosed spaces like bathrooms and bedrooms due to prolonged use of gas heaters or poor ventilation -which can lead to suffocation or even death. In addition, many individuals struggle to secure their homes during long absences, such as vacations or travel, making them vulnerable to burglary.

This project offers a comprehensive technical solution that integrates environmental sensing, facial recognition, and occupancy simulation, all controlled via a dedicated mobile application. The system allows users to monitor their home in real time, control who is allowed access, and receive instant alerts in case of any danger. It reflects a harmonious integration of hardware and software to provide a smart, user-friendly platform that enhances daily safety and peace of mind.

Figure 1.1: Home safety & security system

Problem Statement

Although various security technologies exist in the market, most focus solely on theft prevention, overlooking critical environmental risks that may pose a direct threat to human life. One of the main issues is the dangerous drop in oxygen levels in enclosed areas due to extended use of gas heaters or insufficient ventilation -especially in winter. Such cases can result in unconsciousness or death without an effective alert system. In addition, unauthorized access to homes remains a concern, especially in the absence of intelligent solutions that restrict entry and notify owners instantly.

Furthermore, most homes lack systems that simulate occupancy during extended periods of absence, increasing the risk of burglary. Current solutions are often expensive, complex, or lack integration, making them less effective. This highlights the need for a smart, affordable, easy-to-use, and multifunctional system that ensures both human and home safety.

Aims and Objectives

The general aim is to develop an intelligent, integrated alarm system that enhances individual and home safety by combining air quality sensing and facial recognition technologies, all managed through a dedicated mobile application.

There is another specific objective:

1. Air Quality Monitoring: Develop a reliable sensing unit that detects oxygen levels in enclosed spaces and sends real-time alerts via sound, lights, and mobile notifications when dangerous thresholds are reached.
2. User Identification: Implement an advanced facial recognition system using AI models to restrict home access to authorized individuals only.
3. Occupancy Simulation: Build a feature that simulates presence by intelligently turning lights and devices on/off to mimic household activity during prolonged absences.
4. Mobile Application: Design a dedicated app that manages and monitors the system, displays alerts, allows face registration/removal, and provides air quality reports.
5. System Integration: Ensure that all components operate seamlessly within a unified system, with scalability for future upgrades such as motion detection or smart voice alerts.
6. Ease of Use: Develop a user-friendly interface that allows individuals with no technical background to easily interact with and benefit from the system.

7. Learning mobile app design and real time IOT implementation.

Documentation Layout

Chapter 2 (Related Works) discusses previous studies and existing systems related to home safety, air quality monitoring, and smart security. The chapter highlights how these works inspired or influenced the proposed system and concludes with a comparison table that positions the graduation project among existing solutions.

Chapter 3 (Tools and Methodologies) explains the technical execution of the project. It covers the software development using Flutter and Python (Flask, OpenCV), the hardware components (ESP32, sensors, actuators), and the communication setup (MQTT, Firebase).

Chapter 4 (Experimental Solutions) lists the main problems encountered during development, including software integration issues, hardware connectivity, pin selection, and power management, along with the solutions implemented.

Chapter 5 (Implementation and Results) describes the complete system build and testing. It details the three-layer architecture (mobile app, Flask server, ESP32 hardware), the integration process, and presents the final results, including system accuracy and performance validation.

Chapter 6 (Conclusions and Future Work) summarizes the project's achievements, confirming a successful, affordable, and integrated smart home security system. It also outlines a roadmap for future short-, medium-, and long-term improvements and expansions.

Chapter 2: Related Works

This chapter presents a brief review of previous studies and research in the project's domain. It highlights the main approaches and techniques used in similar solutions, supported by scientific sources that emphasize the significance of the project.

In developing our General Safety and Security Home System, we reviewed existing research to understand current solutions and identify areas for improvement. Among these studies:

2.1 Oxygen Level System Development in WSN and IoT-Based Factory [1]

This study presents the design and implementation of a smart monitoring system for oxygen levels in enclosed industrial environments, utilizing Wireless Sensor Networks (WSN) and Internet of Things (IoT) technologies to enhance occupational safety. The system integrates high-precision oxygen sensors with automated fan control units that adjust ventilation based on real-time data, supported by a cloud-based communication platform for remote monitoring and control. Field testing in an operational factory environment showed a remarkable improvement in oxygen concentration—up to 82%—and a low data loss rate of just 1.67%, confirming the system's efficiency and reliability, although the original context is industrial, the applied principles and technologies form a robust basis for adaptation to home environments, especially in confined spaces such as bedrooms and bathrooms. These features are directly relevant to our project's objective of ensuring smarter and safer residential living conditions.

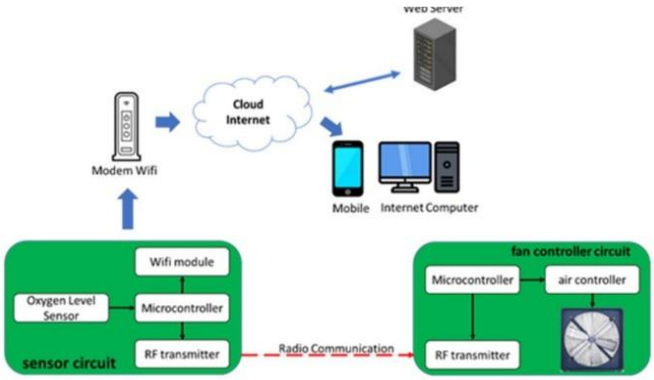


Figure 2.1: The overall system design

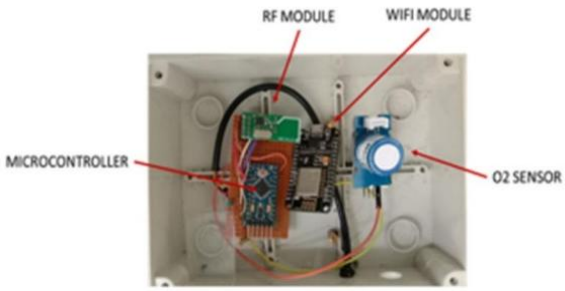


Figure 2.2: The oxygen level sensor 1

2.2 real-time monitoring of carbon dioxide and indoor air temperature using arduino in an effort to maintain indoor air health

[2]

This study focuses on the development of an embedded system for real-time monitoring of carbon dioxide levels and indoor air temperature, aiming to maintain indoor air quality and promote public health. The system is built using an Arduino Uno R3 board, an MG-811 CO_2 sensor, and a DHT11 temperature sensor, supported by an ESP8266 Wi-Fi module, enabling remote data transmission to a central monitoring platform. Field testing conducted in a classroom environment demonstrated the system's effectiveness in accurately detecting environmental changes, providing real-time data that can be utilized to improve ventilation and reduce health risks associated with the accumulation of harmful gases, while the original context of the research focuses on educational environments, the methodology and technologies are directly transferable to residential settings. In our home safety and security system, similar components and monitoring principles can be applied to prevent dangerous accumulations of carbon dioxide, especially in confined areas like bedrooms and bathrooms. This integration not only enhances the system's responsiveness to environmental risks but also supports the broader goal of maintaining a healthy indoor atmosphere as part of an intelligent home safety infrastructure.

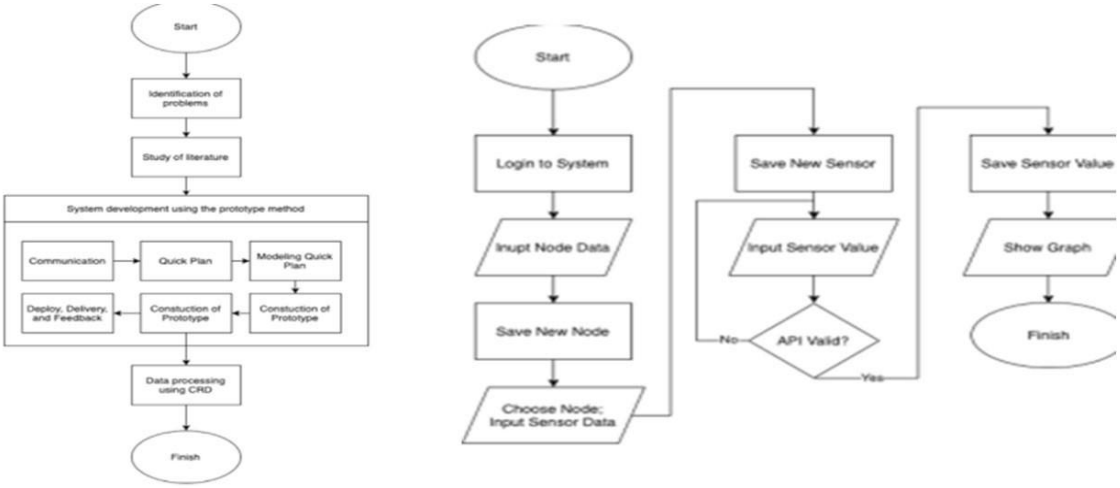


Figure 2.3: System architecture and research frame work

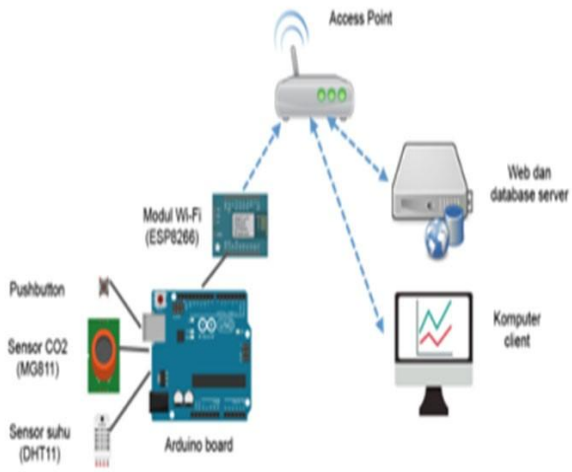


Figure 2.4: System Overview

2.3 Real-Time Monitoring of Indoor Air Quality with Internet of Things-Based E-Nose [3]

This study presents the development of a real-time indoor air quality monitoring system based on an Internet of Things (IoT)-enabled electronic nose (e-nose), designed to detect a wide range of air pollutants in enclosed environments. The system incorporates a suite of gas sensors capable of measuring compounds such as carbon dioxide (CO_2), carbon monoxide (CO), nitrogen dioxide (NO_2), and particulate matter (PM10), along with

environmental parameters like temperature and humidity. These sensors are connected via a wireless communication module, allowing continuous data transmission to a centralized platform for real-time analysis and remote supervision. The system is engineered to be low cost and easy to deploy, making it highly suitable for integration into various indoor settings. Field evaluations confirmed its effectiveness in detecting fluctuations in air quality and enabling timely responses to maintain healthy indoor conditions. While the research primarily targets general indoor environments, the technologies and approaches it applies offer strong potential for residential applications. Integrating similar sensor arrays and IoT frameworks into our home safety system reinforces its ability to ensure a safe, breathable environment especially in enclosed rooms such as bathrooms and bedrooms, thereby aligning directly with the broader objectives of our project.

2.4 Designing a Visitor Verification System Using Facial Recognition Technology via CCTV Cameras

[4]

This study presents the design and implementation of a real-time visitor authentication system utilizing facial recognition technology integrated with CCTV cameras. The system employs a Jetson Nano board and a webcam to capture facial images, which are then processed using deep learning algorithms. In the preprocessing phase, facial data with seven identifiable features are collected. If four or more features are detected, the image is considered to represent a person, and the system proceeds to match the visitor's face with stored user data using 81 feature vectors. The system also includes logging functions that record visitor images, the number of visitors, and visit times, enhancing the security of access control mechanisms. Performance evaluations demonstrated an average detection speed of 6.5 frames per second and an accuracy rate of 86.3%, indicating the system's effectiveness for real-time facial authentication. The methodologies and technologies employed in this research are directly applicable to residential settings. Integrating similar facial recognition systems into our home security project can enhance the ability to monitor and control access to the premises, particularly in identifying and verifying visitors at entry points. This aligns with our project's objective to create a smarter and safer home environment through advanced surveillance and authentication technologies.

Figure 2.5: The structure of hardware

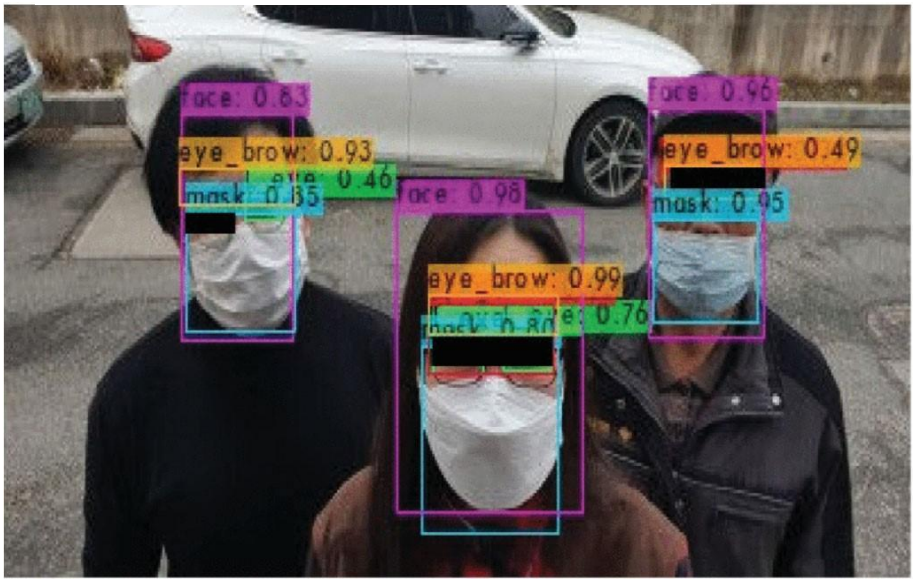


Figure 2.6: Screen that detected the faces of three visitors at the front door.

2.5 Home Security System Using Face Recognition and IoT Technology [5]

This study presents the development of a facial recognition-based visitor management system designed to enhance security through real-time authentication and monitoring. The system utilizes a camera setup to detect human faces, capturing photographs of individuals standing in front of the system. These images undergo preprocessing to ensure clarity and accuracy, employing filtering techniques such as linear, median, and average filtering to remove noise and errors. The processed images are then analyzed using the DLIB algorithm, which extracts facial features and generates feature vectors that remain consistent despite transformations like scaling and rotation. The system compares these vectors against a centralized database of registered users to authenticate identities. In cases where a match is not found, the system initiates a visitor option that assists in scheduling appointments with the desired personnel. The implementation of this system demonstrates its capability to reduce security breaches and streamline visitor management processes without human intervention. The integration of advanced facial recognition technology and real-time monitoring contributes significantly to enhancing the security, efficiency, and reliability of visitor authentication in various environments, including commercial buildings and residential complexes.

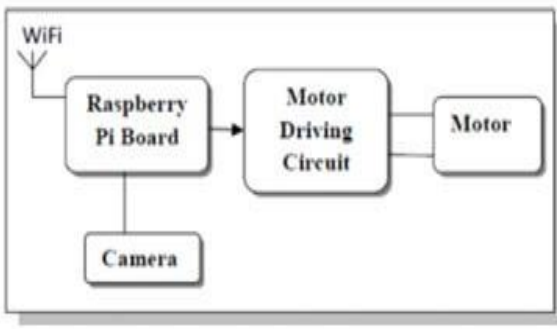


Figure 2.7: Face Recognition-Based Motor

Figure 2.8: Face Recognition Training Interface Control System Diagram.

2.6 Smart Lighting Automation for Enhanced Security and Convenience [6]

This study presents a smart home automation system represented by Philips Hue, which utilizes wireless communication technologies and mobile applications for intelligent lighting control. The system consists of programmable smart bulbs that can be monitored and adjusted remotely in real time, allowing users to create customized lighting scenes that simulate occupancy during absences, thereby enhancing security and comfort, while the system primarily focuses on improving user experience and energy efficiency, the principles and technologies used in such home automation systems provide a strong foundation for a general home safety and security system, including features that simulate presence to prevent intrusion and increase home safety.

2.7 Effortless Control and Automation for Your Home [7]

This study presents the TP-Link HS110 Smart Plug as a practical example of integrating Internet of Things (IoT) technologies into everyday home appliances. The device enables users to remotely control and schedule the operation of electrical devices via a mobile application, while also monitoring energy consumption in real time. The smart plug supports voice control and automation scenarios, enhancing convenience, energy management, and overall home efficiency, although the product is mainly designed to optimize energy usage and provide user-friendly remote control, the underlying technologies, such as remote monitoring, scheduling, and smart control, align well with the broader vision of developing a comprehensive home safety and security system, where appliance control can contribute to simulating occupancy and preventing hazards like overheating or electrical misuse.

Figure 2.9: Kasa Smart Wi-Fi Plug with Energy Monitoring

2.8 Smart Home Security Solution to Deter Burglars with Simulated Light [8]

This study presents the FakeTV device as an innovative solution for enhancing home security through occupancy simulation. Utilizing a built-in computer and super-bright LEDs, FakeTV replicates the dynamic lighting effects of a real television, including scene changes, color shifts, fades, and on-screen motion. These light patterns are projected to mimic the presence of someone watching TV inside the home, thereby deterring potential burglars who often avoid occupied residences. The device is energy-efficient, consuming less power than a standard nightlight, and features a built-in light sensor and timer that automatically activate the unit at dusk for a preset duration. Available in models simulating various TV sizes, FakeTV offers a cost-effective and user-friendly approach to home

security, making it suitable for use during evenings out, vacations, or in unoccupied properties, while FakeTV is primarily designed to simulate occupancy and deter burglaries, the underlying principles and technologies offer a strong foundation for adaptation in residential settings. Integrating such devices into a comprehensive home security system can enhance the illusion of presence, thereby increasing the overall safety and security of the home environment.

Figure 2.10: Model FTV-11

2.9 Related Work Comparison

Table 1.1: Comparison of Projects with Graduation Project.

Project Name	Estimated Cost	Main Tools	Program Language	Problem/Idea Addressed	Automation Level	Home Applicability
GENERAL SAFETY AND SECURITY HOME SYSTEM (our project)	Moderate	ESP32, O_2/CO_2 sensors, Camera, Relay	C++	Home safety and air quality monitoring	High	Yes
Oxygen Monitoring in Factories	Moderate	WSN, Oxygen Sensors, Arduino	C	Industrial oxygen level control	Partial (fan control)	Adaptable
CO_2 and Temperature Monitoring with Arduino	High	Arduino Uno, SCD30 CO_2 sensor, Temp sensor	C++	Indoor air quality and temperature monitoring	Low	Yes
IoT E-Nose for Air Quality	Moderate	ESP32-S3, NH_3/C_2H_4 Sensors, Pumps, Gas Chamber	C++	Toxic gas detection and air quality improvement	Medium	Yes

Facial Recognition via CCTV	High	Camera, AI module, Processor	Python	Visitor verification and access control	Medium-High	Yes
IoT-Based Smart Home Automation	Moderate	Arduino, Sensors, Relay	C++	Basic smart home automation (light/temp control)	Medium	Yes
Philips Hue Smart Lighting System	Moderate	Smart Bulbs, App, Hue Bridge (Hub)	Any (via REST API)	Smart lighting control and customization	Medium	Yes
TP-Link Smart Plug (HS110)	Low	Smart Plug, Mobile App	Any (via REST API)	Remote control and power monitoring of home devices	Medium	Yes
FakeTV Burglar Deterrent Device	Very Low	LED lights, Light sensor, Timer	No programming required	Simulates presence to deter intruders	None	Yes

Chapter 3: Tools and Methodologies

This chapter details the systematic approach and the specific tools employed to develop the General Safety and Security Home System. It provides a comprehensive overview of the project's lifecycle.

Building upon the foundational research outlined in the previous chapter, this section details the practical execution of the project. It describes the rationale behind the selection of specific hardware components, software libraries. Development methodologies are also justified to provide a clear understanding of the project's construction.

The following sections will elaborate on the specific tools, technologies, and step-by-step processes that were adopted to bring the project from concept to a functional prototype.

3.1 Mobile App and Face Detection-Recognition system

The client-facing component of our system is a cross-platform mobile application, developed to provide users with a seamless and intuitive interface for managing their home security. The core development technology selected for this task was the Flutter framework, chosen for its ability to create high-performance, native-quality applications for both Android and iOS from a single codebase. This approach significantly streamlined development and ensured a consistent user experience across different devices.

The development environment was established using Visual Studio Code as the primary code editor, equipped with the official Flutter and Dart extensions to enable powerful features such as code completion, debugging, and the essential Hot Reload functionality for rapid iteration.
A critical phase of the project involved setting up a stable and verified development workspace. The Flutter SDK was installed and its installation was validated using the flutter doctor command, which confirmed all necessary dependencies were met. The development workflow consisted of creating a new Flutter project, connecting a physical Android device for testing via USB debugging, and successfully deploying and running the initial application build. This foundational setup, documented in the steps below, was crucial for the subsequent implementation of the application's features, including the integration of the face detection and recognition module.
[see Appendix A: Table A.1 for more detailed definitions]

3.1.1 Setting Up an Integrated Flutter and VS Code Development Environment

Step 1: Download and install Visual Studio Code **[9]**

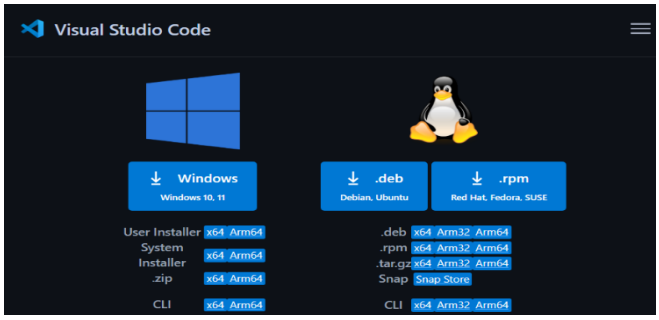


Figure 3.1: Starting vs code Installation

Step 2: Install Flutter SDK **[10]**

A- Depending on what operating system you are using.

Figure 3.2: Starting Flutter Installation

B- Extract it to a permanent folder **C:\src\flutter** on Windows
C- Add the **C:\src\flutter** folder path to your system's PATH environment variable.
D- Open command prompt and run:

```
C:\Users\Home>flutter --version
Flutter 3.35.3 ? channel stable ? https://github.com/flutter/flutter.git
Framework ? revision a402d9a437 (5 weeks ago) ? 2025-09-03 14:54:31 -0700
Engine ? hash 672c59cfa87c8070c20ba2cd1a6c2a1baf5cf08b (revision ddf47dd3ff) (1 months ago) ? 2025-09-03 20:02:13.000Z
Tools ? Dart 3.9.2 ? DevTools 2.48.0
```

Figure 3.3: Starting vs code Installation

flutter doctor shows any missing requirements (Android SDK, platform-tools, Xcode, etc.). All these requirements were checked and completed, as documented with screenshots showing that there are no remaining issues.

```
C:\Users\Home>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[√] Flutter (Channel stable, 3.35.3, on Microsoft Windows [Version 10.0.19045.6332], locale en-US)
[√] Windows Version (10 Pro 64-bit, 22H2, 2009)
[√] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
[√] Chrome - develop for the web
[√] Visual Studio - develop Windows apps (Visual Studio Build Tools 2022 17.14.16)
[√] Android Studio (version 2024.3.2)
[√] VS Code (version 1.104.3)
[√] Connected device (4 available)
[√] Network resources

• No issues found!
```

Figure 3.4: Flutter Doctor Results – System Check

Step 3: Install Flutter and Dart Extensions in visual studio extensions.

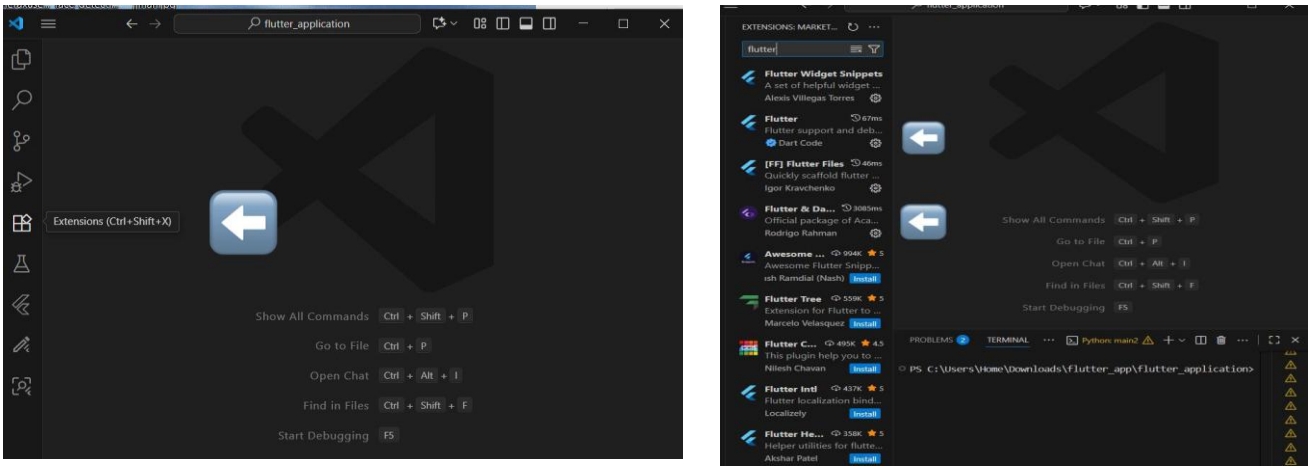


Figure 3.5: Integrating Flutter and Dart in VS Code

Step 4: Create and Run Your First Flutter Project

Step5: Connecting Device and Running Flutter App

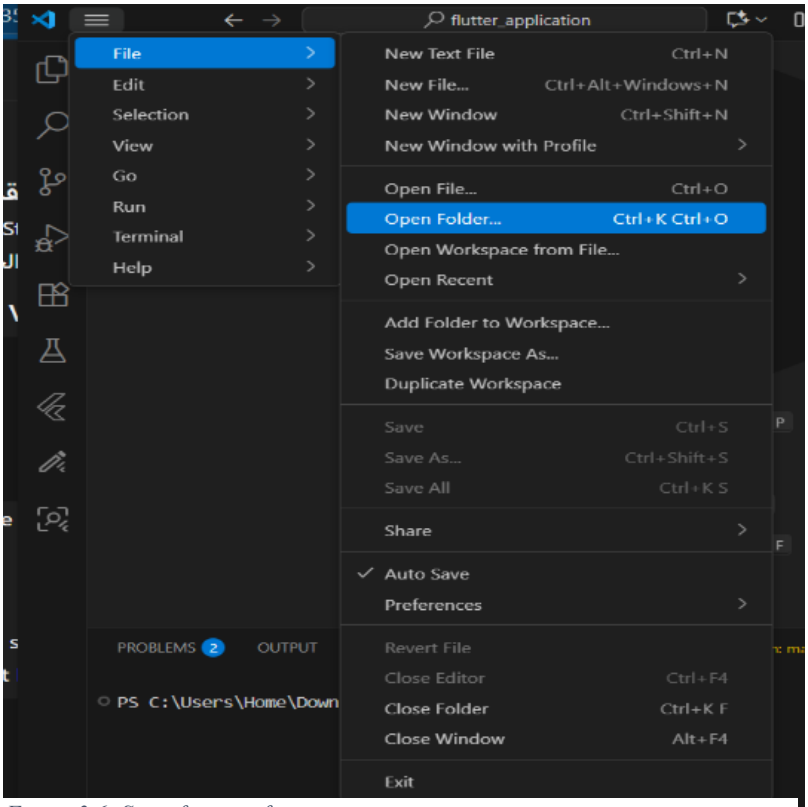


Figure 3.6: Steps for your first app creation

A- Connect your phone to the laptop via USB and make sure USB Debugging is enabled on the phone (in Developer Options).

B-Open the Terminal in VS Code and run the command Flutter Devices to verify that the device is recognized.

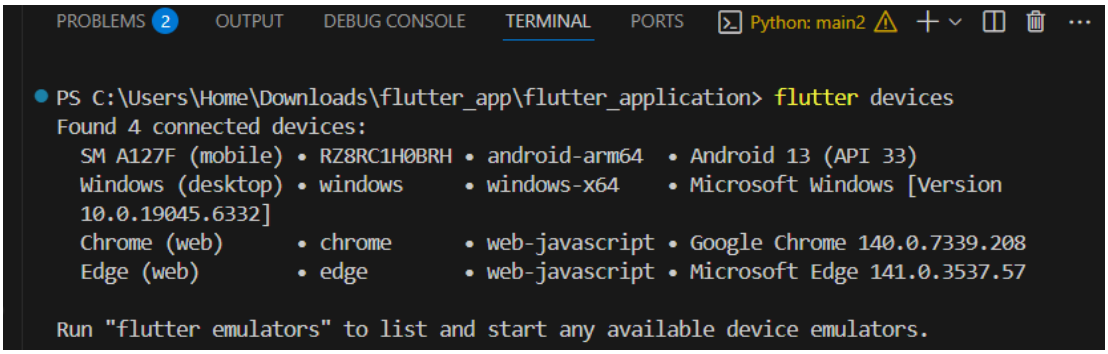


Figure 3.7: Flutter Device Detection Results

C-launch the app directly on the device, run.

```
flutter run
```

Figure 3.8: Flutter command for running the app

3.1.2 Libraries and Packages: Definitions and Descriptions

Software libraries and packages are essential components in modern software development, providing reusable sets of functions, classes, and modules that simplify the implementation of complex functionalities. Instead of writing code from scratch, developers can leverage these pre-built tools to accelerate development, enhance reliability, and ensure consistent performance across applications.

In this project, both Python and Dart libraries are integrated to implement a hybrid system that connects software functionality with physical hardware.

The Python backend provides the core logic for facial recognition, data processing, server communication, and hardware connectivity, while the Dart-based Flutter frontend manages user interaction, image capture, and communication with the backend. The following subsections describe the major libraries and packages employed in both environments.

3.1.2.1 Python Libraries
[11]

The Python backend utilizes several essential libraries to handle web services, image processing, data security, and hardware communication:

Flask [12]:
A lightweight and flexible web framework used to build the RESTful API that enables communication between the mobile application and the backend server. It manages HTTP requests, routes, and responses efficiently, making it ideal for embedded or IoT applications.

Os [13]:
A standard Python library that provides tools for interacting with the operating system, including creating directories, managing files, and accessing environment variables. It ensures that user data and image files are stored and retrieved in an organized and secure manner.

cv2 (OpenCV) [14]:
An open-source computer vision library used for image manipulation, preprocessing, and enhancement. In this project, it handles face detection, cropping, resizing, and contrast adjustment, which are crucial for achieving consistent recognition accuracy.

face_recognition [15]:
The face_recognition library provides an accessible Python interface to dlib's advanced computer vision capabilities, enabling robust face detection and recognition without requiring

machine learning expertise. It serves as the foundation for face-based authentication systems by handling the complex tasks of locating faces in images and converting them into comparable numerical representations.

The library's default detection method uses Histogram of Oriented Gradients (HOG) features combined with a Linear Support Vector Machine (SVM). Images are converted to grayscale, then divided into cells where gradient orientations are calculated and compiled into histograms. These histograms are normalized within overlapping blocks to ensure lighting invariance, creating a comprehensive feature descriptor. A sliding window scans the image at multiple scales, with each window's HOG features evaluated by the pre-trained SVM classifier—trained on millions of face/non-face examples—to determine if it contains a human face.

For recognition, the library employs a modified ResNet-34 convolutional neural network trained using triplet loss metric learning. This network converts detected face regions into 128-dimensional embedding vectors through hierarchical feature extraction—from basic edges to complex facial structures. Crucially, the model learns to position face embeddings in a semantic space where Euclidean distance indicates similarity: same-person faces are clustered together while different-person faces are separated. This enables identity verification by comparing unknown face encodings against a database of known encodings using simple distance calculations, typically with a match threshold around 0.6.

More information about the detection and recognition models shown below.

Table 3.1: Additional Information on Detection and Recognition models.

	Detection (Default: HOG+SVM)	**Recognition (ResNet Deep Metric Learning)**
Goal	Find *where* faces are.	Find *who* the face belongs to.
Output	A cropped picture showcasing the detected face.	A 128-dimensional vector (encoding).
Core Model	HOG features + Linear SVM classifier.	A modified ResNet-34 CNN.
Training Method	Supervised learning on "face"/"not-face" image patches.	Deep Metric Learning using Triplet Loss on face identity triplets.

NumPy [16]:
A core library for numerical computations in Python. It allows efficient manipulation of image matrices and supports the mathematical operations required for encoding and comparing face data.

werkzeug.utils.secure_filename [17]:
Ensures that uploaded filenames are safely formatted to prevent path traversal and security vulnerabilities. It contributes to the secure storage of user images.

Pickle [18]:
Used for serializing and deserializing Python objects such as dictionaries. In this system, it stores and retrieves hashed passwords in a persistent format.

Hashlib [19]:
Provides secure cryptographic hashing functions. It is used to hash user passwords using the SHA-256 algorithm, ensuring that sensitive authentication data is not stored in plain text.
Together, these libraries establish a secure and efficient backend capable of processing facial data, verifying user identities, and managing communication between the application and the server.

paho.mqtt.client (Paho MQTT) [20]:
An open-source client library that implements the MQTT (Message Queuing Telemetry Transport) protocol -a lightweight publish/subscribe messaging system designed for low-bandwidth and high-latency networks. In this project, it enables real-time communication between the backend and IoT devices such as the ESP32 microcontroller. Through MQTT topics, the backend can send control signals (e.g., to open or close doors) and receive sensor data (e.g., gas levels or motion alerts) efficiently and with minimal delay.

google.oauth2.service_account and google.auth.transport.requests [21]:
These libraries are part of Google's official authentication suite, designed to manage secure access to Google Cloud services through OAuth 2.0. The service_account module allows the backend to authenticate using a service account key, while Request facilitates secure token exchange over HTTPS. Within this project, they provide the foundation for enabling secure communication with cloud services, such as Firebase or Google Cloud Messaging, ensuring that notifications and data exchanges are authenticated and protected.

json [22]:
A standard Python module that provides methods for encoding and decoding data in the JavaScript Object Notation (JSON) format. It is used throughout the project for serializing communication between the client and the server -particularly in sending structured responses, exchanging user information, and configuring MQTT payloads. JSON ensures lightweght and human-readable data transfer across all communication layers.

requests [23]:
A high-level HTTP library that simplifies sending and receiving HTTP/HTTPS requests in Python. It handles RESTful communication with external APIs and services, such as cloud platforms or notification systems. In this project, it supports backend communication with third-party services - including potential integrations for push notifications and remote monitoring. Its simplicity and reliability make it ideal for server-to-server data exchange in IoT-based systems.

3.1.2.2 Flutter packages and Dart Libraries [24]

The Dart-based Flutter frontend provides a cross-platform interface that enables users to interact seamlessly with the backend and underlying IoT systems. The following libraries support the main application functionalities:

dart:convert [25]:
A core Dart library that facilitates the encoding and decoding of data in JSON format. It allows structured communication between the Flutter frontend and the Python backend through HTTP requests and responses.

dart:io [26]:
Provides access to file and network input/output operations. In this project, it is responsible for handling locally stored image files and interacting with device-level resources.

flutter/material.dart [27]:
The primary Flutter UI framework that provides prebuilt widgets and design elements adhering to Google's Material Design standards. It ensures a consistent, user-friendly interface across Android and iOS platforms.

image_picker [28]:
Enables the application to capture photos directly from the device camera or select existing images from the gallery. This feature allows users to register and verify their identities visually.

image_cropper [29]:
Provides image editing capabilities that allow users to crop and adjust their photos before submission, ensuring that only the relevant portion (i.e., the face) is used for recognition.

http [30]:
A lightweight networking library used for making RESTful API calls. It connects the frontend to the Flask backend by sending and receiving data for operations such as registration, login, dataset updates, and control commands.

dart:developer [31]:
A core Dart library that provides diagnostic and logging utilities for debugging and performance analysis. In this project, it is used to log key application events, errors, and communication states, which aids in monitoring the system's behavior during development and testing. Its integration ensures better traceability and easier troubleshooting during the app lifecycle.

firebase_core [32]:
The foundational package required for initializing and integrating Firebase services within a Flutter application. It ensures that all Firebase modules, such as messaging or authentication,

are properly configured before use. In this system, it serves as the entry point for connecting the mobile app to Firebase services for future enhancements like cloud notifications and analytics.

firebase_messaging [33]:
A Flutter plugin that enables push notifications using Firebase Cloud Messaging (FCM). It allows the application to receive real-time alerts and updates, even when running in the background. This feature is planned for future implementation to notify users about security events, access logs, or system alerts generated by the backend or IoT devices.

flutter_local_notifications [34]:
A cross-platform plugin that enables the display of local notifications on a user's device. Unlike cloud-based notifications, it operates directly within the app to inform users of system events, task completions, or alerts triggered locally (e.g., successful authentication or device connection). It complements Firebase Messaging by ensuring timely notifications even without internet connectivity.

firebase_options.dart [35]:
An automatically generated configuration file that stores Firebase project credentials, such as API keys, project IDs, and sender IDs. It ensures secure and accurate initialization of Firebase services within the Flutter environment. This file is generated during Firebase setup using the FlutterFire CLI and is essential for consistent and secure integration between the app and Firebase's cloud infrastructure.

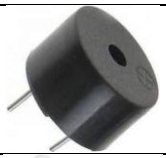
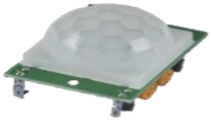
3.2 Hardware Purchase, Activation and simulation

3.2.1 Hardware component characteristics

This section details the selection, procurement, and initial validation of the hardware components that constitute the physical layer of our home security system. Components were specifically chosen based on their functional requirements, compatibility with the ESP32 microcontroller, availability, and budget constraints. The ESP32 serves as the central processing unit that coordinates all sensors and actuators. Following procurement, each component underwent individual functional testing and integration verification using breadboard prototypes and the Arduino IDE. This systematic approach ensured all hardware operated correctly before proceeding to final circuit design and housing assembly.

The table below lists all key components, their images, prices, and specifications. Each part was tested individually with the ESP32 to verify operation before final assembly.

Table 3.2: System Hardware Components' Overview

Component	Component Image	Price	Key Parameters
ESP32-WROOM		7.5JD	Dual-core 240 MHz, 520 KB RAM, 4 MB Flash, 3.3 V operating voltage
2-Channel Relay Module		1.50JD	5 V input, supports 10 A, 250 V AC or 30 V DC
Buzzer		0.50JD	5 V operating voltage
LEDs		0.50JD	2–3.3 V, 10–20 mA current
Servo Motor SG90		2.79JD	5 V, 180° rotation angle
Jumper Wires		0.5JD	Female-to-Female AND Male-to-Female 10 pcs (5×10 cm, 5×20 cm)
Resistors (330 Ω&10k Ω)		0.05JD	330 Ω,10000 Ω
Gas Sensor (MQ-2 / MQ-135)		1.49JD	5 V, analog output
(HW-416-B) motion sensor PIR		0.99JD	3.3V, DIGITAL OUTPUT, Detection Range: 3–7meter, Detection Angle: 110°, Delay Time: 5–300 sec, Current Consumption: 50 µA

			(standby), ~10 mA (active), Warm-up Time: ~60 sec

ESP32 DevKIT (WROOM)
[see appendix B: Table B.1 and B.2 for more details]

The ESP32 DevKit V1 (WROOM-32) is one of the most popular development boards based on the ESP32 microcontroller by Espressif. It features a dual-core processor running up to 240MHz, with built-in RAM and Flash memory for storage.

The board supports both Wi-Fi and Bluetooth connectivity, making it ideal for IoT (Internet of Things) projects. It provides 30 GPIO pins that can be programmed as inputs, outputs, or communication interfaces such as I2C, SPI, UART, and PWM.
It is powered via a Micro-USB port, which is also used for programming through environments like Arduino IDE. ESP has various pins; each designed for a multitude of functions.

GPIO stands for General Purpose Input/Output. These are pins on a microcontroller such as the ESP32 or Arduino, and their function can be controlled through programming. A GPIO pin can act as an input, allowing the microcontroller to read signals from external components like buttons, touch sensors, or temperature sensors. It can also function as an output, enabling the microcontroller to send signals to control devices such as LEDs, relays, or motors.

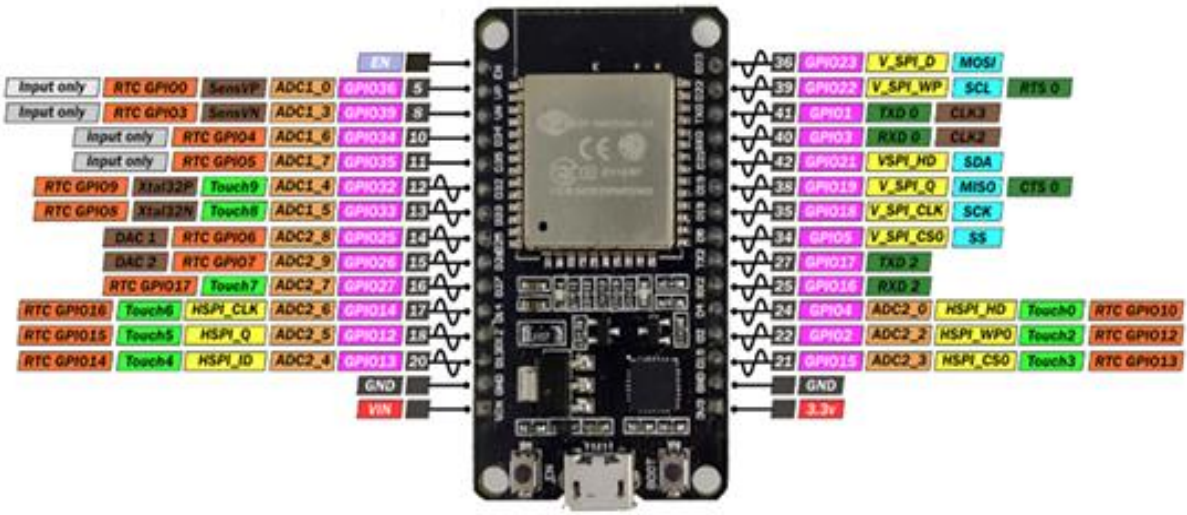


Figure 3.9: Graphical representation of ESP pins and other attached components.

LEDs [36]

To connect an LED to the ESP32, one of the GPIO pins is configured as an output pin. The positive leg of the LED (anode) is connected to a GPIO pin, such as GPIO23, while the negative leg (cathode) is connected through a resistor (typically 330Ω) to GND. The resistor limits the current flowing through the LED to prevent damage.

When the GPIO pin is set to HIGH in the code, the LED turns on; when it is set to LOW, the LED turns off. However, if the LED operates at a higher voltage than 3.3V (for example, a 12V lamp or a high-power LED strip), a Relay Module should be used. The relay provides electrical isolation and allows the ESP32 to safely control higher-voltage circuits without damaging the microcontroller.

In this project, the LED is used as an alert indicator. When motion is detected outside the house, the LED will turn on to signal that movement has been detected, serving as a warning or notification that there is activity outside even when no one is home.

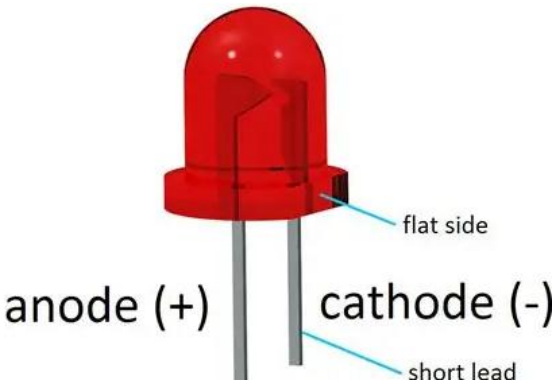


Figure 3.10: LED graphical presentation.

Active Buzzer [37]

To connect a buzzer to the ESP32, one of the GPIO pins is used as an output pin. The positive terminal of the buzzer is connected to a GPIO pin such as GPIO18 or GPIO23, while the negative terminal is connected to GND. A resistor is not used with the buzzer because the current passing through it is relatively low, and the component is designed to operate directly at 3.3V or 5V without the need for current limiting, unlike an LED.

For an active buzzer, simply sending a HIGH signal will turn the sound on, and a LOW signal will turn it off. However, if you need to operate a buzzer that runs on higher voltage or a separate power source, a Relay Module should be used. The relay provides electrical isolation, allowing the ESP32 to safely control higher-power buzzers without risking damage to the board.

In this project, the buzzer serves as an alert system triggered by the MQ-2 Gas Sensor. When the sensor detects smoke or flammable gas, the ESP32 activates the buzzer to emit a warning sound, alerting nearby individuals of potential fire or gas leakage inside the house.

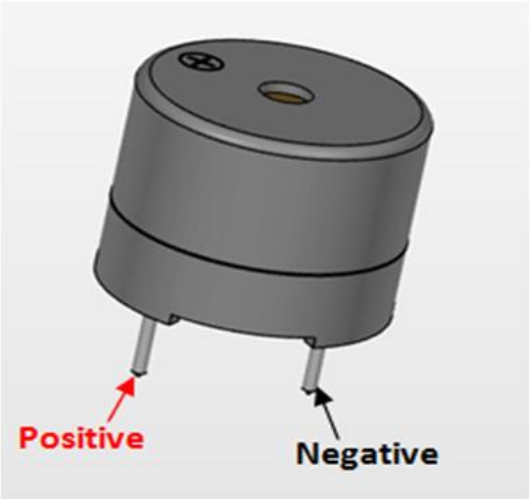


Figure 3.11: Active Buzzer graphical presentation

Servo Motor SG90 [38]

To connect an SG90 servo motor to the ESP32, three wires are used:

- The brown wire (GND) connects to the GND pin on the ESP32.
- The red wire (VCC) connects to the 5V pin to power the servo.
- The orange wire (Signal) connects to one of the GPIO pins (such as GPIO23) to send the control signal.

The servo operates by receiving a PWM (Pulse Width Modulation) signal from the ESP32, where the pulse width determines the rotation angle, typically ranging from 0° to 180°. A resistor is not used because the servo requires higher current than a resistor would allow, and it already includes internal circuitry to regulate current.

A relay module is also not used, since a relay only supports simple ON/OFF control, whereas a servo motor needs continuous and precise PWM signals to control its position -something a relay cannot provide.

It is also important to connect the GND of the servo and ESP32 together to ensure stable and accurate signal communication.

In this project, the SG90 servo motor is used to control door access. After the user's identity is verified and access permission is granted through the mobile application, the ESP32 sends a control signal to the servo to rotate and unlock or open the door automatically.

Figure 3.12: servo motor SG90 graphic presentation

MQ-SENSOR (MQ-2) [39]

The MQ-2 Gas Sensor is widely used for detecting flammable gases and smoke, such as LPG, propane, methane, hydrogen, and smoke. The sensor has four main pins: VCC for power, GND for ground, AO for analog output, and DO for digital output, along with a VR potentiometer to adjust the digital signal threshold. To connect it to an ESP32, connect VCC to 5V, GND to GND, and AO to any ADC pin (such as GPIO32) to accurately read the gas concentration. The AO pin provides a variable analog signal proportional to the amount of gas in the air, allowing continuous monitoring and precise measurement.

The DO pin can be used for HIGH/LOW alerts according to the VR setting. The potentiometer sets a trigger point so that when the gas concentration exceeds this threshold, DO changes state (e.g., LOW for an Active LOW sensor), and when the gas concentration is below the threshold, DO remains in its default state (HIGH for Active LOW). For example, assuming an Active LOW sensor, when the gas concentration is low → DO = HIGH (safe), and when the gas concentration is high → DO = LOW, which can be used to trigger an LED, Buzzer, or Relay as an alert. The VR potentiometer can be adjusted by turning it until DO activates at the desired gas concentration. Once set, whenever the gas exceeds the threshold, DO changes automatically to signal HIGH/LOW.

The sensor requires an initial preheat time of 20–30 minutes for accurate readings, and only a few minutes for subsequent uses. For safety, no external resistor is needed when using AO, as the ESP32 can read the signal directly, and a relay should not be connected directly to the sensor because it only outputs a weak signal, not enough to drive devices. When using DO, a pull-up or pull-down resistor is recommended to stabilize the digital signal. In this project, the MQ-2 sensor will be used to detect the presence of smoke or flammable gases in the environment (for example, smoke caused by fire or gas leaks). When smoke or gas is detected, the sensor will send a signal to the ESP32, which will then activate the buzzer as an alert to warn about the detected smoke or dangerous gas.

Pull-up resistor: A resistor connected between a pin and the positive voltage (VCC) to ensure the pin reads HIGH by default when no other source is driving it. It prevents floating signals that could cause false readings.

Pull-down resistor: A resistor connected between a pin and ground (GND) to ensure the pin reads LOW by default when no other source is driving it. It also prevents floating signals and stabilizes the input.

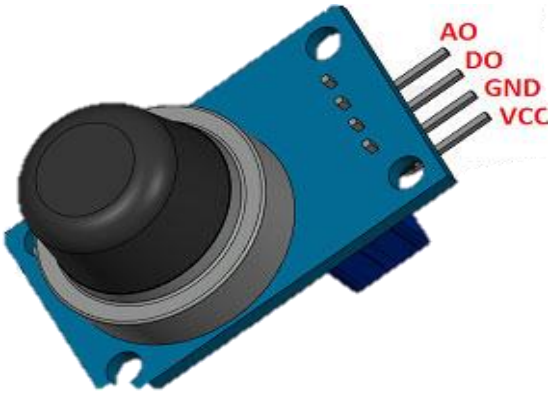


Figure 3.13: MQ-SENSOR graphical presentation

RELAY MODEL [40]

Sending a LOW signal from ESP32 will switch the relay between COM and NO, while HIGH returns it to its default state. Active LOW is useful to reduce load on the ESP32 when multiple relays are connected in a circuit.

Relay pins:

1. VCC → 5V power supply.
2. GND → Ground.
3. IN → Control signal from ESP32 (Active LOW).
4. COM, NO, NC → Switching terminals for the device:
    - COM: Common terminal.
    - NO (Normally Open): Current flows only when the relay is active.
    - NC (Normally Closed): Current flows by default, stops when relay is active.

Connection to ESP32:

- VCC → 5V (external power recommended if current is high).
- GND → ESP32 GND.
- IN → any digital pin; writing LOW on this pin will activate the relay.
- COM & NO → connect the device you want to control.

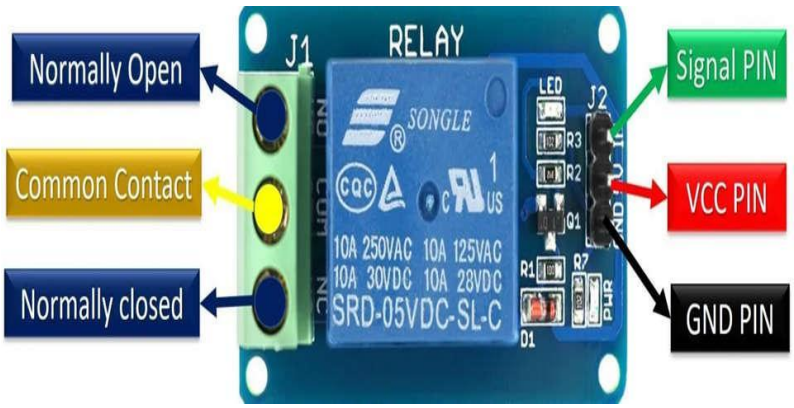


Figure 3.14: Relay graphical presentation.

HW-416-B PIR Motion Sensor [41]

PIR Motion Sensor is widely used for detecting human motion in security systems, automatic lighting, or smart home applications. The sensor has three main pins: VCC for power, GND for ground, and OUT for digital output, along with two onboard potentiometers to adjust detection range (sensitivity) and delay time.

To connect it to an ESP32, connect VCC to 5V, GND to GND, and OUT to any digital GPIO pin to detect motion events. The sensor provides a digital HIGH signal when motion is detected, and LOW when no motion is detected (or vice versa depending on the trigger mode).
The detection range potentiometer can be adjusted by turning it until the sensor detects motion within the desired distance (typically 3–7 meters). The delay time potentiometer sets how long the output remains HIGH after detecting motion (5–300 seconds).

The sensor requires an initial warm-up time of approximately 60 seconds after power-on for stable operation. For reliable readings, a pull-up or pull-down resistor can be used on the OUT pin to prevent floating signals that may cause false triggers.

In this project, the HW-416-B sensor will be used to detect motion in the monitored area. When motion is detected, the sensor will send a digital signal to the ESP32, which can then activate a Yellow LED and Blue LED as an alert.

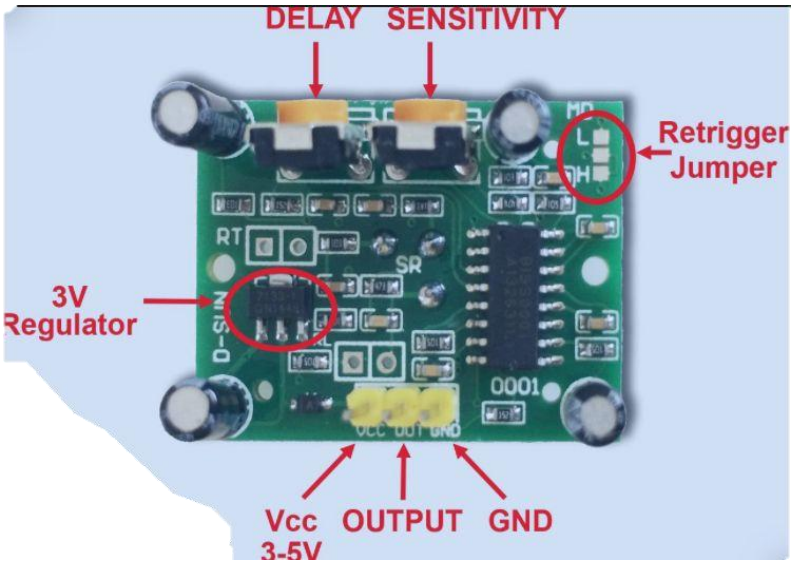


Figure 3.15: graphical presentation of the PIR motion sensor.

3.2.2 Setting Up and Running the ESP32 using Arduino IDE

Step 1: Download and install the Arduino IDE **[42]**

Figure 3.16: Arduino IDE Version

Step 2: Install ESP32 Board Package

1-Go to File → preference → add the needed URL **[43]**

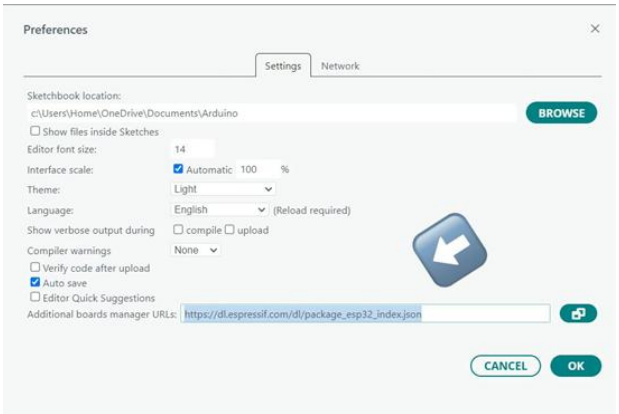


Figure 3.17: Arduino IDE Preferences Configuration

2-Open Tools → Board → Boards Manager, search for ESP32, and install it.

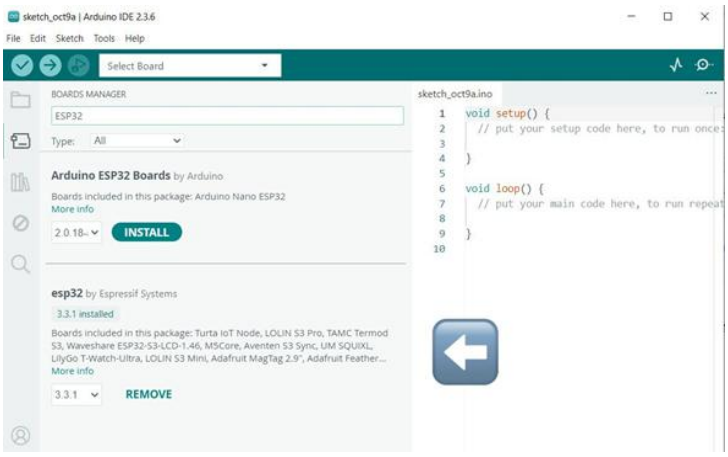


Figure 3.18: Installed ESP32 Board Version in Arduino IDE

Step 3: Select the ESP32 Board and Port

1-Connect the ESP32 to your computer using a USB cable (Type c).
2- go to Tools → Board and select your board model *(*ESP32 Dev Module*)*.
3-choose the correct COM port from Tools → Port →choose (COM3).

3.3 Mobile App and Hardware Connection

At the stage of establishing real-time communication between the mobile application, backend server, and IoT hardware devices, two main communication technologies were carefully evaluated: Firebase and MQTT. Both provide mechanisms for message delivery and synchronization, yet their architectures and performance characteristics differ significantly.

After evaluating several factors that will be discussed in chapter five, MQTT was selected as the primary communication protocol for handling real-time messaging between the ESP32 microcontroller and the Flask backend server. This choice ensures low latency, efficient bandwidth usage, and reliable bidirectional communication necessary for timely device control and status updates.

Meanwhile, Firebase remains integrated into the system with a more focused role -it handles push notifications and user alerts within the Flutter application. This hybrid approach allows the system to leverage the strengths of both technologies: MQTT for real-time hardware interaction, and Firebase for seamless mobile-level user communication.

3.3.1 MQTT Installation and Setup

MQTT stands for Message Queuing Telemetry Transport. It is a lightweight messaging protocol designed to enable fast and efficient communication between devices, especially in Internet of Things (IoT) networks or any system requiring low-bandwidth, real-time data exchange. **[44]**

The protocol uses a Client-Server architecture:

- Broker (Server): Receives messages from devices and distributes them to other devices subscribed to specific topics.
- Client (Device/Application): Any device or application that sends or receives data.

MQTT uses a Publish/Subscribe model rather than direct device-to-device communication, reducing network load and simplifying data management.

Mosquitto Broker [45]

Mosquitto is one of the most popular open-source MQTT brokers**.** It is compatible with Windows, Linux, and macOS and supports all MQTT features: Quality of Service (QoS)**,** retained messages, and authentication. This is the type of broker used in our project.

3.3.1.1 Installing Mosquitto on Windows [46]

Red: indicating the Command or PowerShell commands.

Green: indicating program responses and functions.

Step 1: Download Mosquitto **[47]**

Step 2: Install Mosquitto broker:

Select the Windows version (usually a .exe file).

During installation, choose to add Mosquitto Service to run it automatically as a server.

Step 3: Run Mosquitto Broker:

Open Command Prompt or PowerShell and type:

net start mosquitto

Or for temporary manual execution:

mosquitto

If running correctly, you will see a message like:

1625: New client connected from 127.0.0.1.

Step 4: Testing Mosquitto:

Open Terminal 1 and subscribe to a topic:

mosquitto_sub -h localhost -t "test/topic"

Open Terminal 2 and publish a message:

mosquitto_pub -h localhost -t "test/topic" -m "Hello MQTT"

→ You should see the message "Hello MQTT" appear in Terminal 1.

3.3.1.2 ESP32: MQTT Connection Steps [48]

Step 1: Connect to WiFi

- Use SSID and password
- Wait until WiFi.status() == WL_CONNECTED
- Print local IP for verification

Step 2: Initialize MQTT Client

- Create PubSubClient with WiFi client
- Set server address and port
- Set callback function for incoming messages

Step 3: Reconnect Automatically

- Use a reconnect() function
- Retry connection every 5 seconds until successful

- Subscribe to topic: home/door → Door control

Step 4: Handle Incoming Messages
- Callback function receives topic and payload
  Example: Door: open / close → Servo moves

Step 5: Publish Messages from ESP3
- Oxygen alert: home/oxygen_alert
- Motion alert (PIR sensor): home/motion_alert

3.3.1.3 Flask: MQTT Connection Steps [49]

Step 1: Initialize MQTT Client
- Use paho.mqtt.client.Client("FlaskServer")
- Connect to broker using same IP and port
- Start loop_start() to keep connection active

Step 2: Subscribe to Topics
- home/oxygen_alert → Receive oxygen level alerts
- home/motion_alert → Receive motion detection alerts

Step 3: Handle Incoming Messages
- Use on_message() callback
- Decode payload
- Print or process alerts

Step 4: Publish Commands to ESP32
Door: mqtt_client.publish("home/door", "open") or "close"

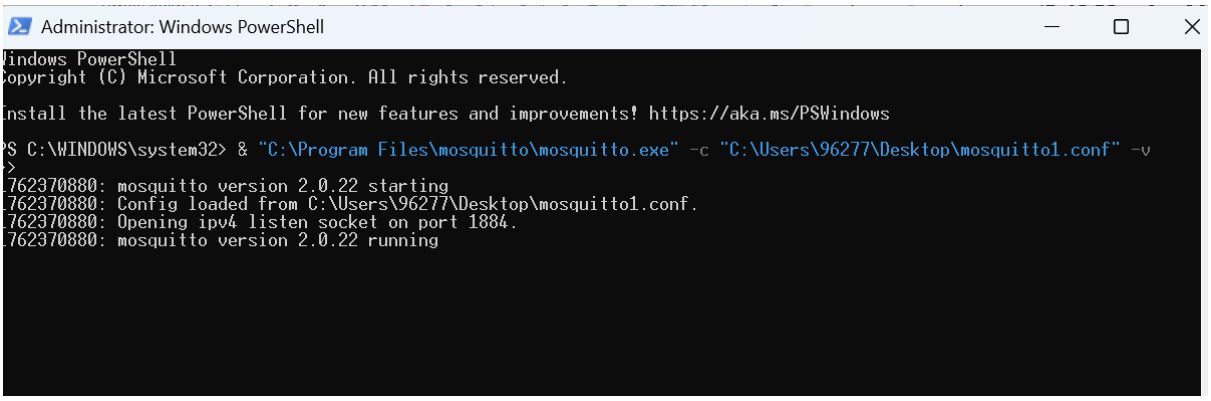


Figure 3.19: Running Mosquitto.

3.3.2 Firebase Installation and Setup

Firebase remains an integral part of the system but with a more specialized role. Firebase is a cloud-based application development platform from Google that provides a comprehensive suite of services for modern app development, including cloud messaging, authentication,

analytics, and hosting **[50].** In this project, Firebase Cloud Messaging (FCM) is used to send real-time push notifications to the Flutter mobile application whenever important events occur, such as:

- Oxygen level alert (Oxygen Alert)
- Motion detected outside the house (Motion Alert)

This hybrid communication structure allows the system to leverage the strengths of both technologies: MQTT for reliable, low-latency device-to-server messaging, and Firebase for user-facing event notifications and cloud-based scalability.

3.3.2.1 Firebase Cloud Messaging (FCM) [51]

FCM is Firebase's service for sending real-time notifications and messages.
Notifications can be sent directly from the server (Flask backend) to devices subscribed to a specific topic.

Key features of FCM:

- Supports background messages and notifications even when the app is closed.
- Allows sending messages to a single device or a group of devices subscribed to a topic.
- Reliable, fast, and works on both Android and iOS.

Setting up Firebase is a simple procedure.
Setup Steps:

1. Create a project in the Firebase Console.
2. Download the serviceAccount.json file for server authentication.
3. Configure FCM with Flask to use the service file for sending messages.
4. Subscribe to a topic in the Flutter app to receive notifications.

3.3.2.2 Flask Backend Integration

Step 1: Authentication with Firebase:

- We used a serviceAccount.json file downloaded from the Firebase Console.
- The file provides credentials for Flask to authenticate with Firebase.

Step 2: Sending Notifications:

- We used the google-auth library to generate an access token.
- The access token is used to authorize HTTP requests to Firebase Cloud Messaging.
- A function send_fcm_notification(title, body) was created to:
- Specify the topic (e.g., "oxygen_alerts").
- Define the notification title and body.
- Send a POST request to FCM to deliver the message

Example Usage in Backend:

```
send_fcm_notification("Oxygen Alert", "Oxygen High!")
```

Whenever the ESP32 publishes a high oxygen alert to the MQTT broker, the Flask backend receives it and immediately sends a notification via FCM.

3.3.2.3 Flutter App Integration

Step 1: Initializing Firebase:

- Firebase is initialized at the start of the app using Firebase.initializeApp().
- Flutter listens for notifications both in the foreground and background.

Step 2: Subscribing to Topics:

- FirebaseMessaging.instance.subscribeToTopic("oxygen_alerts");
- This ensures the app receives notifications for events published to the oxygen_alerts topic.

Step 3: Handling Notifications:

- Foreground: Using FirebaseMessaging.onMessage.listen() to display in-app notifications.
- Background**:** Using a background message handler _firebaseMessagingBackgroundHandler.

Step 4: Local Notifications:

- The Flutter app uses flutter_local_notifications to show notifications on the device.
- This allows users to see alerts even when actively using the app.\

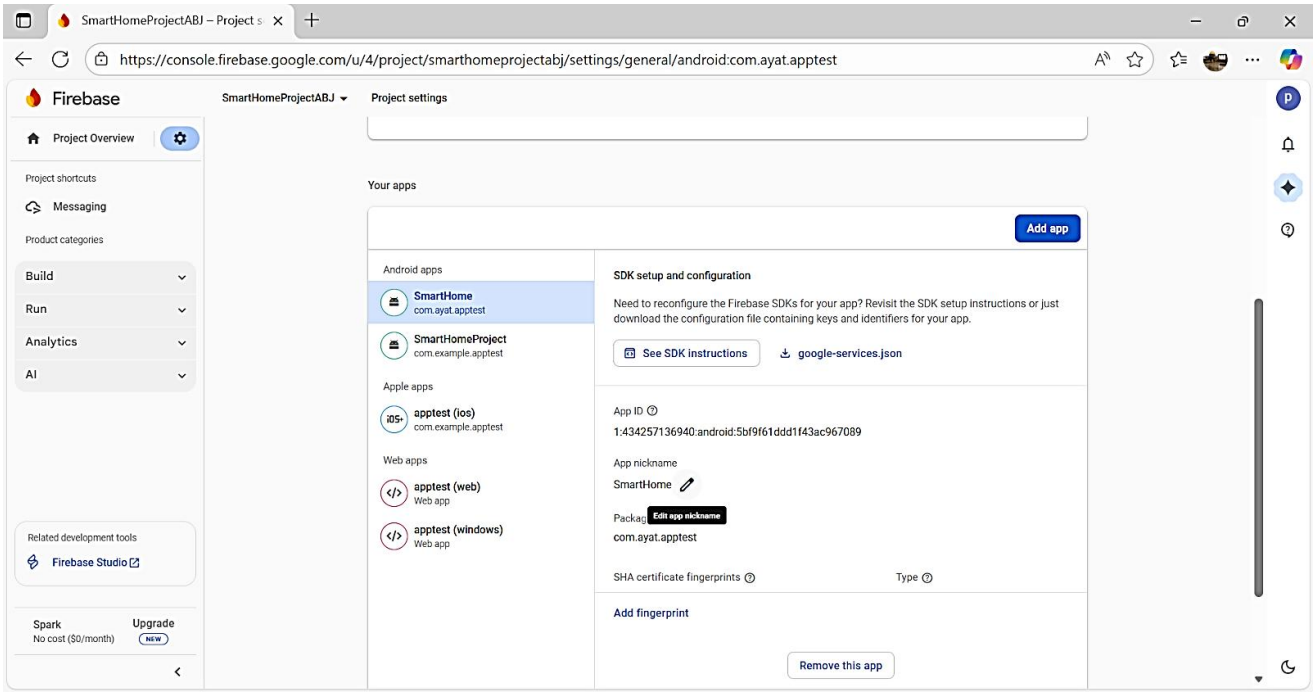


Figure 3.20: Flutter project account and our app.

Chapter 4: Experimental Solutions

The development of the General Safety and Security Home System, while ultimately successful, was not without its obstacles. This chapter details the significant technical, logistical, and implementation challenges encountered throughout the project's lifecycle. It describes the nature of each problem, the analytical process undertaken to diagnose the root cause, and the specific solutions or workarounds that were implemented to overcome them. Documenting these difficulties provides a complete picture of the development process and offers valuable insights for future projects.

4.1 Software Phase

The software implementation phase presented several significant challenges, primarily centered on the integration of a machine learning model into the Flutter-based mobile application. The key difficulties involved optimizing the model for mobile performance, managing device resource constraints, and creating a seamless user experience despite the computational demands of real-time face processing.

Selection of Development Stack
At the initial stages, our team encountered difficulties in selecting an appropriate development stack that could effectively integrate machine learning–based face recognition with a mobile application. Implementing face detection directly within Flutter proved impractical due to limited access to mature machine learning libraries and compatibility issues between AndroidX, OpenCV, and TensorFlow Lite.

Solution:
The team adopted a hybrid approach by implementing the backend using Python and Flask, leveraging Python's robust libraries (OpenCV, face_recognition, NumPy) for image processing and model management. This shift enabled faster prototyping, reduced build complexity, and allowed for easier debugging and fine-tuning of recognition algorithms.

Face Detection and Image Preprocessing
Face detection posed significant challenges, particularly under low lighting, non-frontal angles, and variable image resolutions. Furthermore, inconsistencies in color representation (BGR vs. RGB) affected detection accuracy.

Solution:
The system was improved by converting all images to grayscale, normalizing image sizes, and applying bounding boxes to ensure proper face extraction. These measures enhanced detection precision and consistency during both registration and authentication phases.

Face Encoding and Recognition Efficiency
Encoding facial features into numerical vectors (embeddings) initially resulted in performance delays, especially when processing high-resolution images or multiple users simultaneously.

Solution:
To address this, images were resized before encoding, and duplicate embeddings were filtered out. This optimization significantly reduced computational time and improved recognition speed without sacrificing accuracy.

Image Upload and Format Handling
Transferring images from the Flutter application to the Flask server introduced challenges related to file size, format compatibility (JPEG, PNG), and network interruptions.

Solution:
The system was enhanced using `http.MultipartRequest` with `multipart/form-data` to ensure reliable transmission. The backend also incorporated automatic format validation and conversion to maintain compatibility across devices.

User Interface Responsiveness and State Management
The team faced challenges in maintaining a responsive user interface while the application performed background operations such as image uploads and recognition. In some cases, the interface would temporarily freeze, creating a poor user experience.

Solution:
State management was optimized using Flutter's `setState()` method and asynchronous programming. A temporary "Processing..." indicator was implemented to inform users during operations, ensuring smooth interaction.

Camera and Storage Permission Handling
Differences in Android device settings caused inconsistent behavior in accessing the camera and gallery. Some devices failed to grant permissions automatically, hindering the image capture and upload process.

Solution:
The team implemented runtime permission requests and tested the application on various devices to ensure compatibility and reliability across different Android versions.

Error Handling and User Feedback
Unclear or generic error messages previously caused confusion when server or network errors occurred.

Solution:
A comprehensive error-handling mechanism was introduced on both the Flutter and Flask sides, providing clear, user-friendly notifications such as "Cannot read image" or "Access denied." This improved transparency and usability.

Server Performance and Data Management
As the number of registered users increased, face comparison operations slowed due to the expanding dataset. Additionally, updating or deleting users could cause inconsistencies if embeddings were not properly synchronized.

Solution:
The backend was optimized to store only unique embeddings, and folder structures were organized by username. Whenever user data was updated, embeddings were regenerated to maintain system consistency and recognition accuracy.

4.2 Hardware and Connection Phase

During the implementation of the Home Security System, the team encountered multiple challenges related to hardware integration, circuit design, and device communication. Establishing reliable connections between the various hardware components - including sensors, actuators, and the ESP32 microcontroller -required careful attention to pin selection, power distribution, and wiring accuracy. Moreover, ensuring stable real-time communication between the hardware and the backend introduced additional difficulties related to protocol compatibility, library limitations, and Wi-Fi reliability.

These challenges demanded a deep understanding of the ESP32's architecture, as well as iterative testing and troubleshooting to ensure that each component operated correctly and communicated seamlessly within the system. The following sections outline the key hardware and connection issues faced during development and the strategies adopted to overcome them.

Pin Selection Challenges
During the project implementation, our team realized that selecting the appropriate pins on the ESP32 was a significant challenge. Depending on the ESP32 model we used, pins such as Input-Only, Analog-to-Digital 1 (ADC1), Analog-to-Digital 2 (ADC2), as well as pins (6-11) GPIO that cannot be used for certain applications without affecting the ESP32's performance during operation or training. Therefore, it was crucial for the team to carefully understand the function of each pin and use it correctly, ensuring proper operation of all components and avoiding potential system issues.

Solution:
the team carefully studied the ESP32 Pinout Diagram in detail and learned the function of each pin and what can be connected to it. Afterwards, we prepared a simple Pin Assignment Table, in which we listed all the pins we intended to use for each sensor, actuator, or module.

Power Supply and Wiring Challenges
During the implementation of the Home Security System project, our team faced challenges related to the power supply due to the varying voltage requirements of different components. Some components operated only at 5V, others only at 3.3V, while some could function on

either voltage. This variation posed a practical challenge in distributing power appropriately across the breadboard and ensuring that voltage drops or sudden failures did not occur.
In particular, when we enabled the Wi-Fi on the ESP32, some components, such as the MQ-2 sensor and the MG90S servo motor, became less stable at 3.3V and only operated reliably at 5V. This required careful management of the power distribution to maintain system stability.
During the practical implementation, the hardware team also faced challenges with electrical wiring and connections between components. The system relied on various elements such as the ESP32 microcontroller, servo motors, MQ gas sensor, breadboard, jumper wires, and resistors used for voltage and current regulation. In some cases, wiring mistakes occurred -for example, connecting a signal wire to the wrong pin on the ESP32 or mixing up the 3.3V and 5V power lines -leading to improper functioning of certain components and unexpected behavior in the circuit.

Solution:
To overcome the issues related to power distribution and wiring, our team created a small team-specific chart (Team Pin & Wiring Chart) that included all the connections for each component on the ESP32, specifying the corresponding pins and the appropriate voltage for each component **[Table 5.1].** Additionally, the team conducted practical testing for each component with its assigned connections and voltage to ensure stable operation at the correct voltage.

Sensor Reading Variability Challenge
During the course of our project, our team observed that the sensor itself operates correctly, but the varying operational environment causes readings to differ from one location to another. This natural variation presented a practical challenge for us when determining the appropriate threshold for triggering an alarm or assessing the presence of a hazard.

Solution:
our team ensured that environmental factors, such as ventilation, temperature, and humidity, were carefully considered when calibrating the sensor to maintain system stability and accuracy.

Difficulty in Organizing Wires and Circuits
As the system expanded with the addition of multiple sensors and outputs, our team faced a challenge in organizing wires and circuits, as the wires became tangled and difficult to manage, which could lead to connection errors or maintenance difficulties. In addition, we had more than one component, and each component required a specific voltage, while we only had two 3.3V pins, two 5V pins, and two Ground pins, which was insufficient for all the components.

Solution:
To overcome this challenge, the team initially designed a well-organized layout on the Breadboard.

Component Integration Challenges with ESP32
During the project implementation, our team faced several challenges in connecting different components with the ESP32. Some components did not work directly with the ESP32 without modifying libraries or changing pins. For example, the servo motor did not operate immediately when connected to the ESP32; it required downloading the appropriate servo libraries in the Arduino IDE and configuring the settings accordingly.
Additionally, we encountered another challenge with certain sensors, such as the MQ-2 Sensor, where we had to change its pin from ADC2 (Analog Digital 2) to ADC1 (Analog Digital 1) to ensure compatibility with Wi-Fi operation on the ESP32.

Solution:
Our team conducted a detailed study of the ESP32 pins to identify which pins operate more reliably with Wi-Fi, ensuring stable system performance.

Handling Sudden Failures
Our team observed that the design of certain components, such as LEDs and resistors, was relatively fragile, so that any minor shock or movement could cause them to break or fail. This means that even a small bump or incorrect handling could result in a broken resistor, damaged LED, or disconnection of wiring, including the LED's anode or cathode.

Solution:
the team managed these failures effectively thanks to having spare parts on hand, allowing quick replacement of any damaged components and ensuring the project continued without interruption.

Connectivity Approach and Challenges
Although we used Firebase for notifications in our project, it was not integrated as a core part of the system due to difficulties in connecting the hardware components with the software application. Using Firebase required installing heavy libraries, such as Firebase_ESP_Client, which were relatively large for the ESP32 and placed additional load on the processor. Additionally, maintaining secure SSL connections was challenging because SSL encrypts the data between the device and the application, and the ESP32's limited resources and memory made it difficult to sustain stable connections, especially with heavy libraries.

Solution:
To overcome these challenges, the team opted for MQTT as a lighter, faster, and more efficient alternative, enabling smooth and reliable communication between the ESP32 and the application. Furthermore, since the ESP32 supports only 2.4 GHz Wi-Fi networks, we ensured that the system was connected exclusively to compatible networks to prevent connection losses and maintain stable performance.

Chapter 5: Implementation and results

This chapter presents the practical realization of the proposed home security system, describing in detail the implementation of both the software and hardware components, followed by the evaluation of the system's performance.

The implementation process involved integrating the mobile application, backend server, and hardware modules into a unified framework designed to enhance home safety and user accessibility.

The first section provides an overview of the development environment and the configuration of the tools and technologies used. The subsequent sections illustrate the implementation of the mobile application using Flutter and the backend server using Python (Flask). Additionally, the integration between the ESP32 microcontroller and the backend is discussed, outlining how the various components communicate and function together.

Finally, the chapter concludes with a presentation of the testing results, demonstrating the accuracy and reliability of the system in real-world conditions.

5.1 System Overview

This chapter presents the complete practical development of the General Safety and Security Home System, documenting the journey from architectural design to a fully functional prototype. The system successfully integrates three distinct technological layers into a cohesive security platform: a mobile application built with Flutter for user interaction, a Python Flask server handling facial recognition and system logic, and an ESP32-based hardware module equipped with environmental sensors and motors.

The system architecture follows a three-tier model with distributed processing:

1. **Presentation Tier**: Mobile application built with Flutter framework providing user interaction and real-time monitoring capabilities.
2. **Application Tier**: Python Flask server implementing RESTful API endpoints, facial recognition algorithms, and communication middleware.
3. **Hardware Tier**: ESP32 microcontroller with sensor network including environmental detectors, motion sensors, and motor controls.

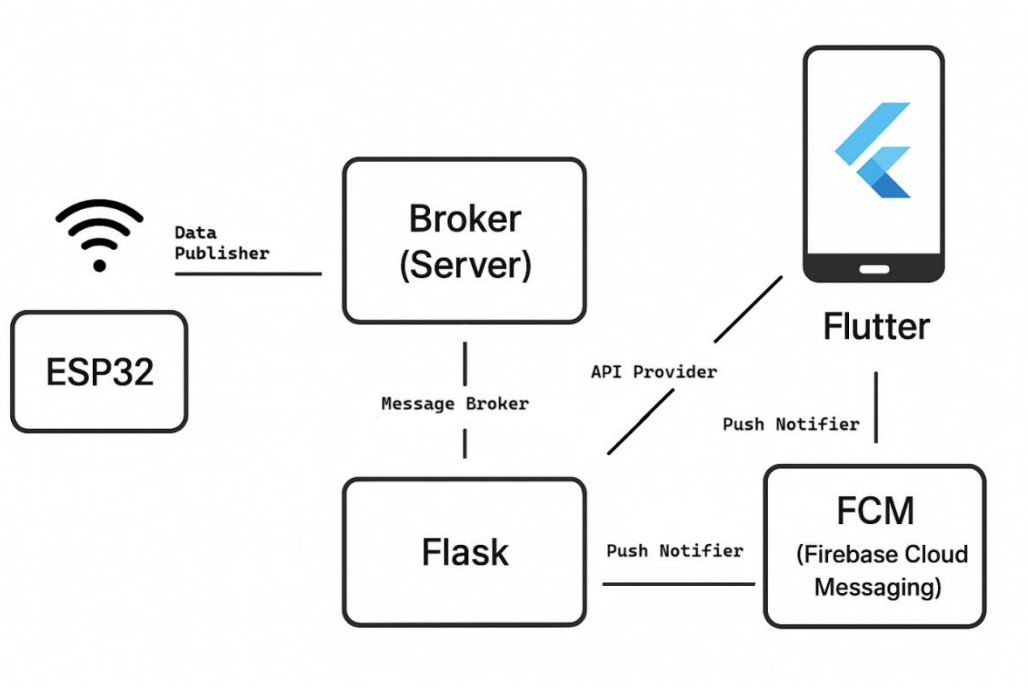


Figure 5.1: Project Overview Diagram

Development followed a structured, phase-based methodology. Each subsystem was designed, prototyped, and validated independently, ensuring reliability and simplifying debugging. This modular approach allowed for parallel progress and incremental testing before final system integration. The completed prototype demonstrates a seamless workflow where real-time home monitoring, secure biometric access control, and immediate environmental hazard alerts operate in unison, validating the core concept of an affordable, integrated smart home security solution.

5.2 Frontend Implementation Using Flutter

The client-facing component is a cross-platform mobile application developed with Flutter, chosen for its single-codebase efficiency. The implementation focused on creating an intuitive, secure interface that guides users through home security management with minimal complexity.

5.2.1 User Interface Design

The mobile application interface was crafted following Material Design 3 guidelines (using Theme in MaterialApp in flutter) with emphasis on intuitive navigation and accessibility. The design process involved multiple iterations with user feedback incorporation:

Navigation Structure:

- Linear workflow from authentication to main dashboard reduces cognitive load as it guides users step-by-step without presenting multiple pathways, which simplifies decision-making and minimizes confusion.

- Clear visual hierarchy established through consistent typography and spacing that intuitively directs user attention to primary actions and information.

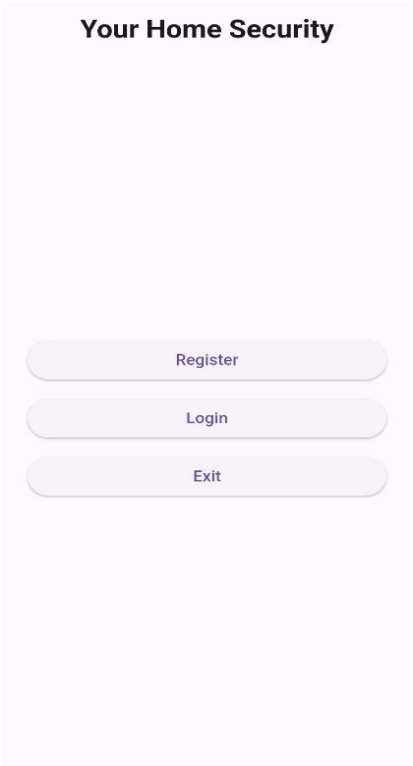


Figure 5.2: The Apps Home screen

5.2.2 Authentication System Implementation

The authentication system implements a two-step verification process that combines facial recognition with password confirmation. This approach balances security with practical usability for home access control, ensuring that only authorized users can operate the security system.

Facial Recognition Phase

The authentication process begins with biometric verification through facial recognition. When a user attempts to log in, the mobile application activates the device's camera to capture a facial image. The captured image is immediately transmitted to the Flask backend as a compressed JPEG file using a multipart/form-data POST request to the /recognize endpoint. This method efficiently handles both the image data and any accompanying metadata in a single transmission.

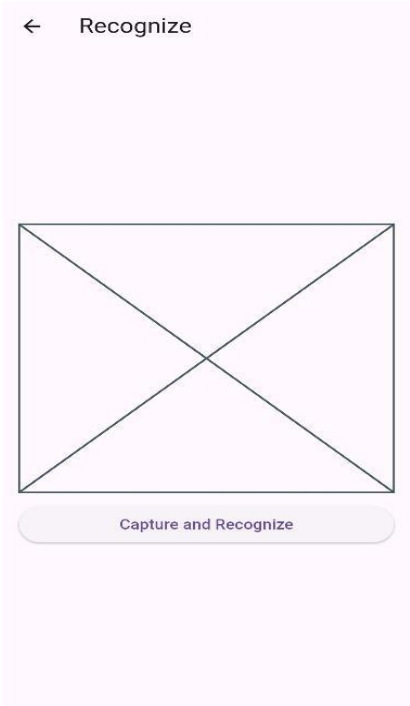


Figure 5.3: The Recognition Page

On the server side, the received image undergoes processing through the facial recognition pipeline. The system first detects and extracts the face region, then converts it into a 128-dimensional numerical encoding that represents the facial features. This encoding is compared against all stored facial encodings in the user database. If a match is found within the configured threshold, the system proceeds to the next verification phase. If no match is found or the image quality is insufficient for reliable recognition, the user receives immediate feedback through the mobile interface (snack-bars) and can attempt recognition again.

```
C:\Users\96277\Desktop\connection app\backend>python server4.py
Notification sent: 200 {
  "name": "projects/smarthomeprojectabj/messages/3022869025802892422"
}

Connected to MQTT for Oxygen alerts, code: 0
 * Serving Flask app 'server4'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.100.15:5000
Press CTRL+C to quit
192.168.100.11 - - [12/Dec/2025 15:17:23] "GET / HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 15:17:49] "GET / HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 15:17:52] "GET / HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 15:18:38] "POST /recognize HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 15:18:44] "POST /verify_password HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 15:18:44] "GET /dataset HTTP/1.1" 200 -
```

Figure 5.4: The Server Responding to Recognition Requests

Password Verification Phase

Following successful facial recognition, the system requires password confirmation to complete authentication. The mobile application presents a password entry dialog specifically for the recognized user, ensuring they understand which account they're accessing. The entered password is transmitted to the Flask backend via a JSON-formatted POST request to the /verify_password endpoint.

The Flask server processes this verification by retrieving the stored SHA-256 hash for the identified user from the passwords.pkl file. It computes the hash of the provided password and compares it with the stored value. This server-side hashing approach ensures that plaintext passwords are never stored or transmitted unnecessarily. If the hashes match, authentication succeeds and the user is granted access to the security dashboard. If verification fails, the user can retry the password or restart the authentication process.

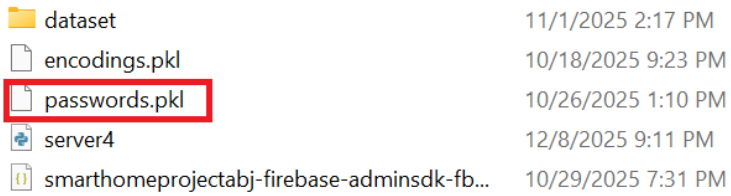


Figure 5.5: Password PKL file

Security Considerations

The authentication system employs several security-conscious design choices that enhance protection without complicating the user experience. The transmission of facial images and passwords occurs over the local network without additional encryption layers, as the prototype operates within a trusted home environment. The server implements basic input validation to prevent malformed requests, and the stateless design requires complete verification for each sensitive operation rather than relying on persistent sessions.

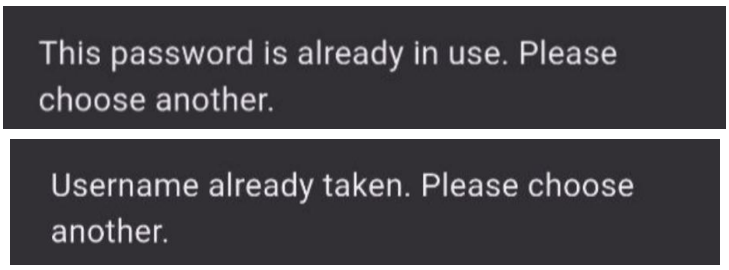


Figure 5.6: Failed Attempts at entering username and password

This two-phase approach creates a practical balance for home security applications. The facial recognition provides convenient biometric access while the password confirmation adds an essential knowledge-based layer of security. Together, they form an authentication system appropriate for controlling physical access points while maintaining usability for household members.

5.2.3 Dashboard and Control Interface

The dashboard serves as the central command center, organized into functional zones.

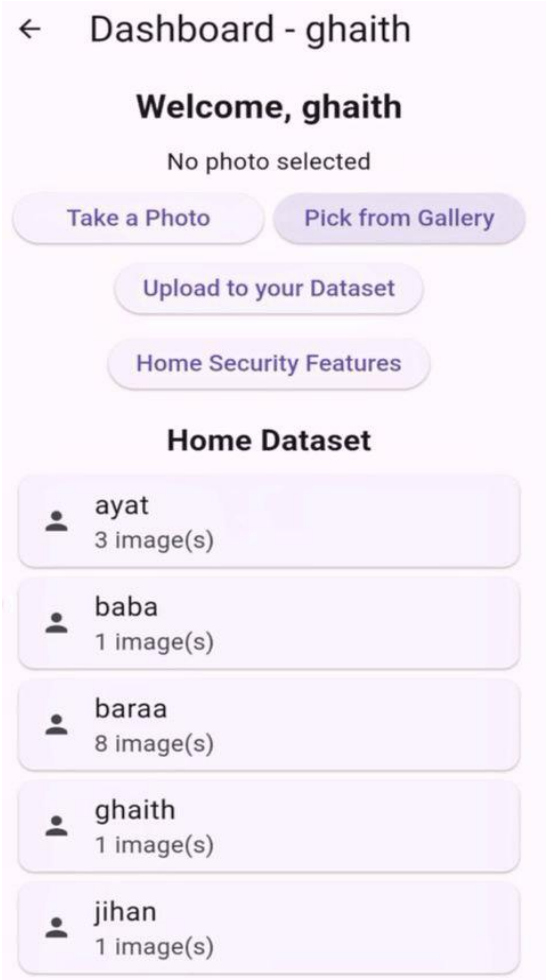


Figure 5.7: A User's Dashboard

Dataset Management Interface:

- Visual representation of registered users with image counts
- Dynamic update: Automatic update to the dataset when a user registers or adds a photo

Security Features Implementation:

- Door Control: Large, clearly labeled buttons for open/close commands with visual state confirmation

- Motion Sensor Management: Individual sensor enable/disable toggle that indicates weather the users are home or not. The system activates motion monitoring only when the **"**Outside Home**"** button is pressed, and deactivates it when "Inside Home" is selected. When active and motion is detected, three LED indicators located inside and outside the home light up sequentially as a visual alert.

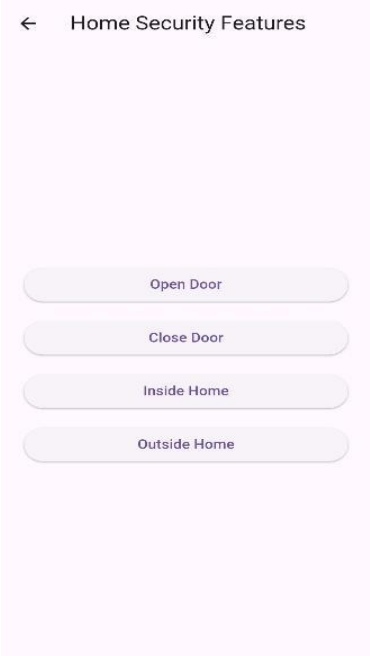


Figure 5.8: The Security Page

5.2.4 Notification and Alert System

The notification system implements multi-channel alerting using Firebase Cloud Messaging otherwise called FCM:

Notification Types and Triggers:

- Warning alerts: Oxygen threshold breaches, Motion detection when the user is not home.
- Confirmation notices: Action completion (door opened/closed), successful commands (user home/not home)

Delivery Mechanisms:

- In-app notifications: Snack-Bar and dialog-based alerts within the application interface
- Group notification framework: Structure prepared for multi-user alert distribution via FCM when activated
- Visual indicators: buzzer and LED activation

With the mobile interface providing user access to security features, the system's intelligence resides in the backend server, which we now examine.

5.3 Backend Implementation

The backend server, implemented in Python using the Flask framework, serves as the central processing hub that orchestrates all system operations, from facial recognition to hardware communication.

5.3.1 Server Configuration and API Design

The backend server implements a lightweight RESTful API architecture using the Flask framework. This architectural style organizes system functionality around resource-based URLs and standard HTTP methods—such as GET for retrieving data and POST for creating resources or triggering actions. All communication between the mobile application and server follows stateless request-response patterns using lightweight JSON data formats, ensuring interoperability, simplicity, and scalability in line with modern web service standards.

Core Server Capabilities:

- RESTful API endpoints for user registration, facial recognition, and hardware control
- Concurrent request handling through Flask's operation mode
- Error handling with appropriate HTTP status codes (200, 400) and JSON-formatted messages

Communication Architecture:

- Dual-protocol support: HTTP/REST for mobile communication, MQTT for hardware integration
- Real-time messaging: Persistent MQTT client connection for immediate hardware control
- Response optimization: Efficient JSON payload design minimizing transmission overhead

5.3.2 Face Recognition Processing

The facial recognition system implements a practical computer vision pipeline using established libraries to provide reliable user identification for home security applications. Built on the face_recognition Python library -which itself utilizes dlib and OpenCV- the system balances accuracy with processing efficiency suitable for real-time authentication scenarios.

Image Processing Pipeline

When a facial image arrives at the Flask server, it undergoes a structured transformation process to extract comparable facial features. The system first validates the incoming image file, confirming it's a supported format and within reasonable size limits for processing. The image is then converted to RGB format, as the facial recognition algorithms expect this standard representation.

Core Processing Steps:

- Face Detection: Histogram of Oriented Gradients (HOG) method identifies facial regions within images (a well-established computer vision technique that analyzes edge directions within image regions to reliably locate facial structures under varying lighting conditions.)
- Image Normalization: Detected faces are cropped, resized to 250×250 pixels, and contrast-enhanced
- Feature Extraction: A pre-trained deep learning model generates 128-dimensional facial encodings
- Quality Assurance: Basic validation ensures only clear, properly oriented faces proceed to recognition

The core detection phase analyzes gradient directions in localized portions of the image to identify facial structures regardless of lighting variations. When a face is detected, the system extracts the corresponding region and prepares it for feature extraction. In cases where multiple faces appear in a single image, only the most prominent or first-detected face is processed for authentication purposes, if the image was added using the gallery, this issue can be solved by cropping the unwanted faces using the cropping feature in the dashboard .

Feature Encoding and Matching

The preprocessed facial image is converted into a compact numerical representation using a pre-trained deep learning model (SVM). This model generates a 128-dimensional feature vector that mathematically represents the unique characteristics of the face, enabling mathematical comparison rather than visual analysis.

Matching Process:

- Encoding Generation: Each face transformed into a numerical vector capturing distinctive features
- Database Comparison: New encodings compared against all stored user encodings using cosine distance
- Similarity Thresholding: Configurable threshold determines match acceptance or rejection
- Result Identification: Closest matching user returned for password verification phase

During user registration, these encodings are stored alongside user credentials in the system's dataset. When a user attempts authentication, their new facial encoding is compared against all stored encodings using cosine distance measurement -a mathematical technique that quantifies similarity between vectors regardless of their magnitude.

Data Management and System Optimization
The facial recognition system implements efficient data handling to maintain performance as the user database grows. Facial encodings are stored in memory during server operation, eliminating repeated disk access during authentication attempts.

Performance Considerations:

- Organized File Structure: User images stored in hierarchical dataset/ directory by username
- Processing Optimization: Image resizing before encoding reduces computational load
- Duplicate Prevention: Registration process prevents identical usernames or passwords in database to prevent confusion in recognition phase

The modular design allows for straightforward updates to recognition algorithms or threshold values as system requirements evolve. This implementation provides a balanced solution for home security applications, offering sufficient accuracy for access control while maintaining the simplicity and transparency needed for educational demonstration and practical deployment in household environments.

Following the examination of biometric processing, we explore how the server communicates with physical hardware components to enable real-time monitoring and control

5.3.3 Hardware Communication using MQTT

The system implements lightweight MQTT communication between the Flask backend and ESP32 hardware, chosen for its low latency and efficient publish-subscribe model ideal for IoT applications.:

MQTT Architecture and Configuration:

- Broker Configuration**:** Local Mosquitto MQTT broker running on the same machine as the Flask server
- Connection**:** Persistent client connection established at server startup
- Topic Structure**:** Simple topic naming for clarity and ease of debugging:
  - Control commands: home/door (for door open/close)
  - Sensor alerts: home/oxygen_alert and home/motion_alert

Message Format and Handling:

- Command Messages: Simple string commands (e.g., "open", "close", "toggle") for direct hardware control
- Alert Messages: Human-readable alert strings (e.g., "Check on your air levels!") published when sensor thresholds are breached
- Message Processing: Direct string comparison in ESP32 firmware for immediate action execution

Communication Workflow:
The system's command flow begins with the mobile application sending requests to a Flask HTTP endpoint, which then publishes corresponding messages to the MQTT broker. The ESP32 subscribes to these topics and performs the required hardware actions based on the received commands. In parallel, the alert flow operates in the opposite direction: when an ESP32 sensor detects an event, it publishes an MQTT message that the Flask server subscribes to, enabling logging and preparing the data for future notification processing.

For overall system reliability, status monitoring is maintained through MQTT's built-in keep-alive mechanism, ensuring that all components remain connected and responsive.

Reliability Features:

- Automatic Reconnection: Both Flask and ESP32 implement reconnection logic for network interruptions
- Connection Verification: Simple publish-subscribe verification at startup
- Error Handling: Basic try-catch around MQTT operations to prevent server crashes
- Local Network Optimization: Configured for low-latency operation within home Wi-Fi environment

Implementation Details:

- Flask Integration: MQTT client runs in separate thread using Paho MQTT library
- ESP32 Integration: PubSubClient library with callback-based message handling
- Broker Setup: Mosquitto running on default port 1884 with localhost access
- Message Routing: Direct topic-based routing without intermediate processing

Current Limitations and Scalability:

- Simple Messaging: Uses basic string messages rather than structured JSON payloads
- No Acknowledgment: One-way command delivery without confirmation feedback
- Local Scope: Optimized for single-home, local network deployment
- Basic QoS: Default quality of service (QoS 0) for most messages

The MQTT implementation delivers minimal communication latency between software and hardware components, enabling near-instantaneous control responses and prompt alert notifications that meet the real-time requirements of home security applications.

5.3.4 Database and Storage Management

The backend implements efficient, file-based data management tailored to the prototype's requirements:

User Data Storage Architecture:

- Directory Structure: Organized folder hierarchy with separate directories for each registered user
- Image Storage: JPEG files named sequentially (username_1.jpg, username_2.jpg) for easy management
- Metadata Management: Pickle-serialized dictionary storing username-password hash mappings
- Facial Encodings: In-memory storage during operation with potential for disk serialization

Data Persistence Implementation:

- Password Storage: passwords.pkl file containing the hashed passwords with automatic loading/saving
- Image Organization: dataset/ directory containing subfolders for each user's facial images
- Temporary Processing: In-memory image manipulation during recognition to minimize disk I/O
- Backup Strategy: Manual backup capability through file system copying

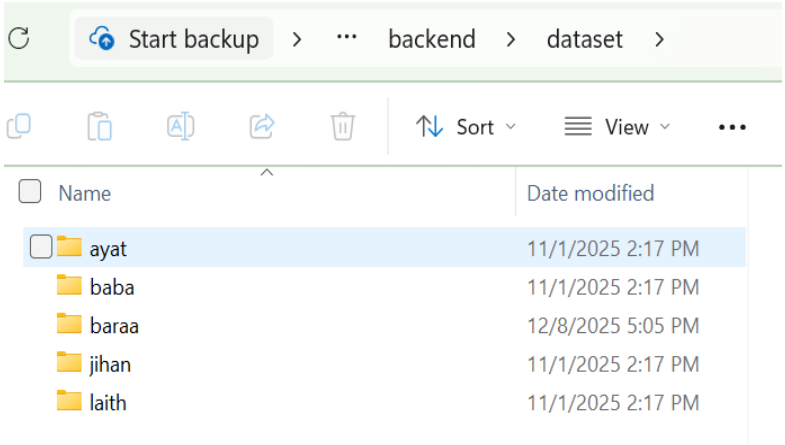


Figure 5.9: Folder Hierarchy for Dataset in the Server

Data Management Operations:

- User Registration: Creates user directory, stores hashed password, saves initial facial image
- Image Addition: Appends new images to existing user directories with sequential naming and gives you the ability to edit the image to ease the load on the detection model.
- Dataset Querying: Filesystem scanning to enumerate users and count their images

- Data Integrity: Automatic folder creation for new users and validation of existing structures

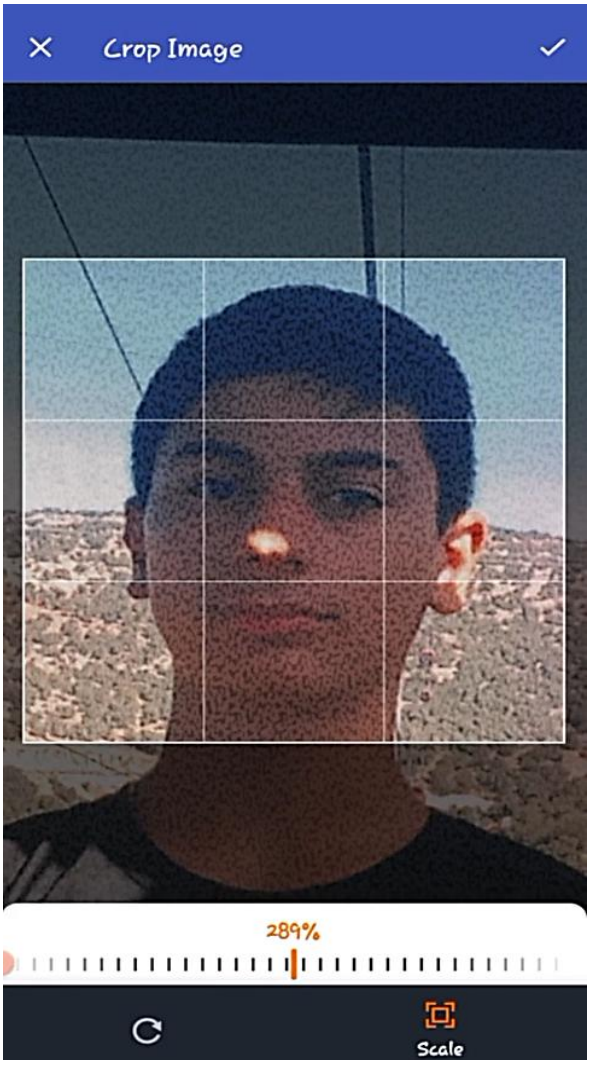


Figure 5.10: Edit on the Photo you want to upload in the dataset from the gallery feature

Privacy and Security Considerations:

- File Permissions: Standard filesystem permissions for data protection
- Data Isolation: Separate storage per user prevents cross-user data access
- Cleanup Procedures: Manual removal required for user data deletion

5.3.5 FCM Push Notifications Implementation using Firebase

The Flask backend includes basic Firebase Cloud Messaging configuration to support future cloud-based alert delivery. This setup establishes the server-side infrastructure for sending push notifications to mobile devices when security events occur. All of the notification actions can be seen from the firebase project website.

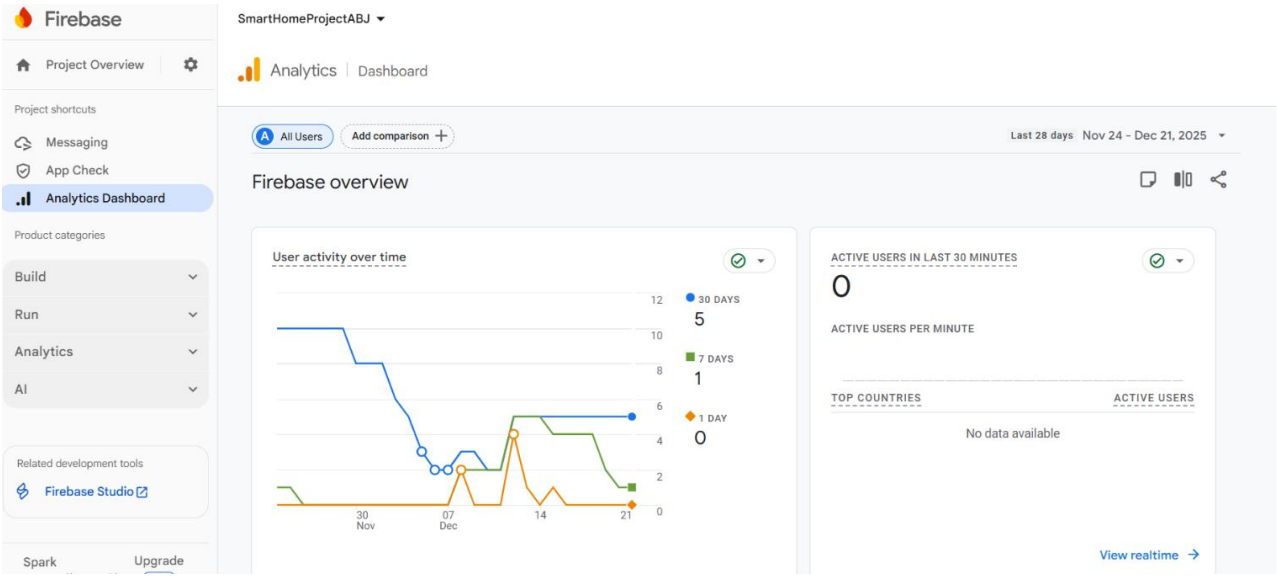


Figure 5.11: The Project's Firebase Analytics Dashboard Showcasing the User Activity

Service Configuration

- Firebase Authentication: Loads serviceAccount.json credentials to ensure secure and authorized API access.
- Token Management: Uses OAuth2 to generate valid tokens required for authenticated Firebase requests.
- API Integration: Configures an HTTP client to communicate safely and reliably with FCM endpoints.

The project dashboards keep tabs on the frequency of the notifications and users over time. This ensures that we have up-to-date information and history of users and their notifications for future data analysis.

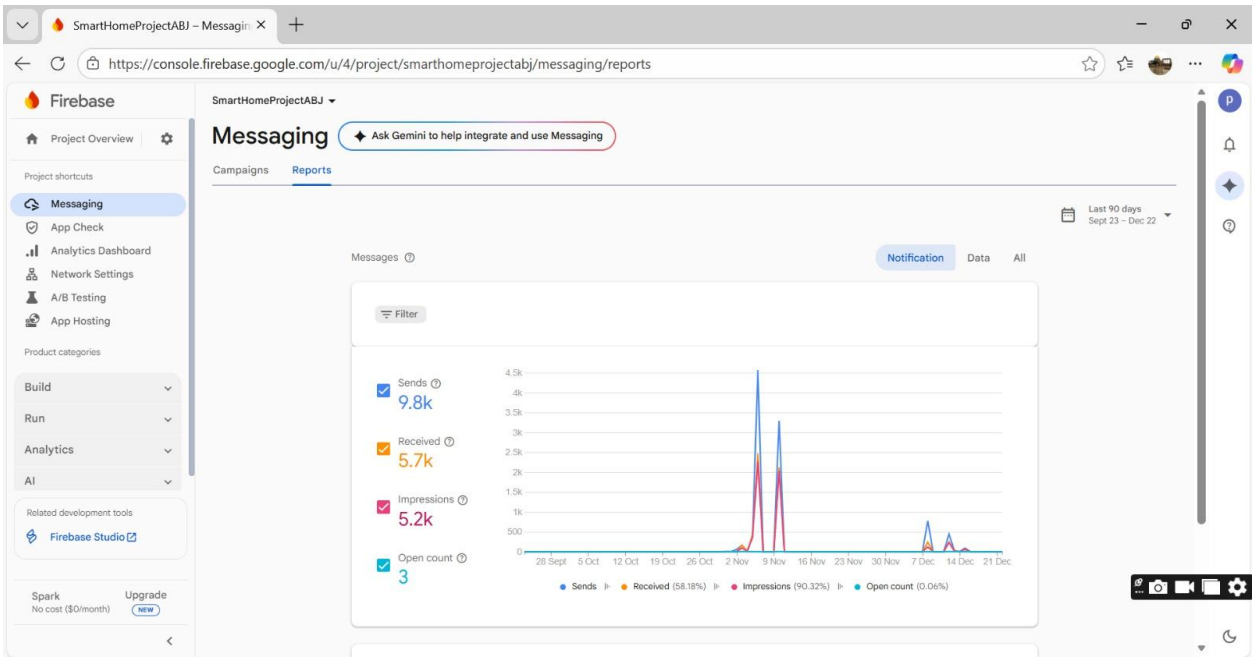


Figure 5.12: FCM Reporting Dashboard

Notification Functionality

- Alert Sending Function: Implements send_fcm_notification() with title and body parameters for flexible and easy alert creation.
- Topic Support: Organized to handle multiple alert categories such as oxygen alerts and motion alerts.
- Payload Formatting: Structures outgoing messages in basic JSON format to meet FCM requirements.
- Error Handling: Wrapped in try-catch logic to manage network or authentication failures gracefully.

Integration with System Events

- MQTT Trigger Integration: Sensor alerts from the ESP32 can directly trigger outgoing FCM notifications.
- Event Categorization: Maps oxygen threshold breaches and motion detection events to the appropriate notification types.

Current Implementation Status

- Configuration Complete: Firebase service account is fully integrated and functional.
- Basic Functionality: Core notification-sending logic is implemented and already testable.
- Future-Ready: The design allows easy scaling into a full notification system.

With the backend computational and communication infrastructure established, we now detail the hardware implementation that brings the system into physical interaction with the home space.

5.4 Hardware Implementation Using ESP32

The hardware implementation followed a structured development methodology that progressed from organized breadboard prototyping to integration within a physical home simulation model. This approach enabled systematic validation at each stage while maintaining flexibility for adjustments and optimizations.

5.4.1 Component Integration and Physical Assembly

The physical implementation began with strategic breadboard organization where components were grouped by function to facilitate testing and debugging. A detailed pin assignment table was maintained throughout development, documenting each connection's purpose and characteristics. This documentation served as the blueprint for both the initial prototype and the final system integration.

Implementation Methodology:

A. Connection Documentation: Comprehensive pin assignment table specifying GPIO functions and interfaces

Table 5.1: The Hardware to Pin Connection

No.	Component	Pin	Connected To (Pin)	Signal Type	Notes
1	**MQ-2 Gas Sensor**	VCC	5V	Power	Operates on 5V only
		AO	10k on GPIO34	Input	Requires 10KΩ resistor
2	**Servo Motor**	VCC	5V	Power	Operates on 5V only
		GND	GND	Ground	
		Signal	GPIO23	Output	
3	**PIR MOTION SENSOR**	VCC	5V	POWER	Operates on 5V only
		GND	GND	Ground	Requires 10KΩ resistor
		Signal	10k on GPIO13	output	
4	**BUZZER**	Cathode (-)	GND	Ground	

		Anode (+)	GPIO22	Output	Active High/Digital output
5	**YELLOW LED**	Cathode (-)	GND	Ground	Requires 330Ω resistor Digital output
		Anode (+)	NO switch	Output	
	BLUE LED	Anode (+)	330Ω on GPIO17	Output	
		Cathode (-)	GND	Ground	
	RED LED	Anode (+)	330Ω on GPIO18	Output	
		Cathode (-)	GND	Ground	
6	RELAY	VCC	3.3V	POWER	ACTIVE LOW. When the relay is activated, NO connects to COM, allowing current to flow to the load. Use this terminal when you want the load to be powered only when the relay is ON.
		GND	GND	GROUND	
		Vin	GPIO26	INPUT	
		Switch Comm	5V	Switching point	
		Switch NO	330Ω YELLOW LED ANODE	Switch / Load	

B. Digital Simulation: Circuit modeling prior to physical implementation to validate connections **[52 53]**

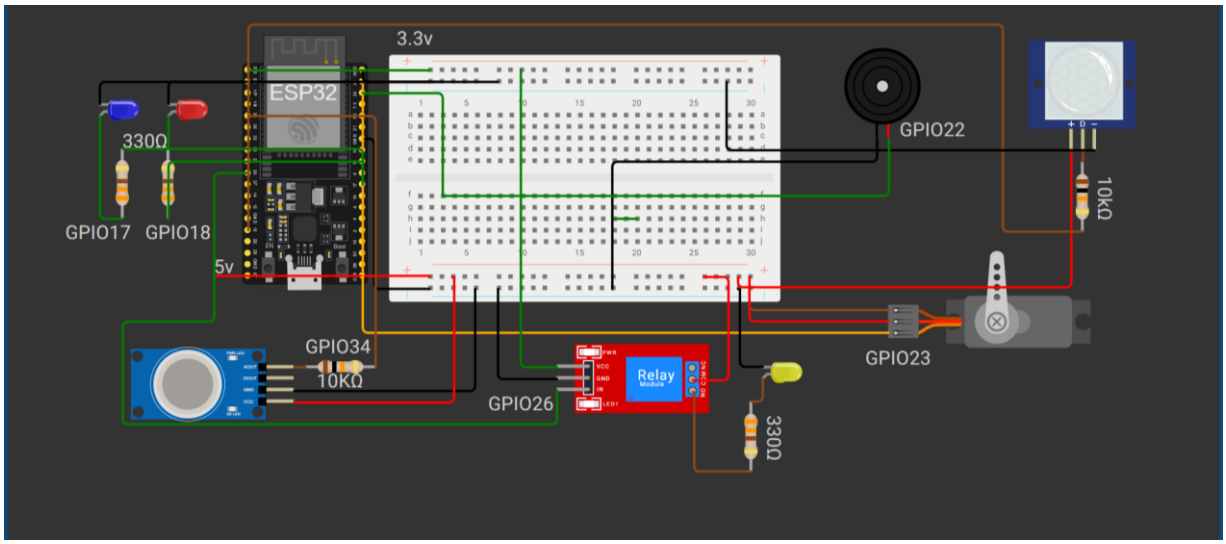


Figure 5.13: Hardware Assembly Simulation

C. Breadboard Prototyping: Initial circuit assembly with components arranged logically by function

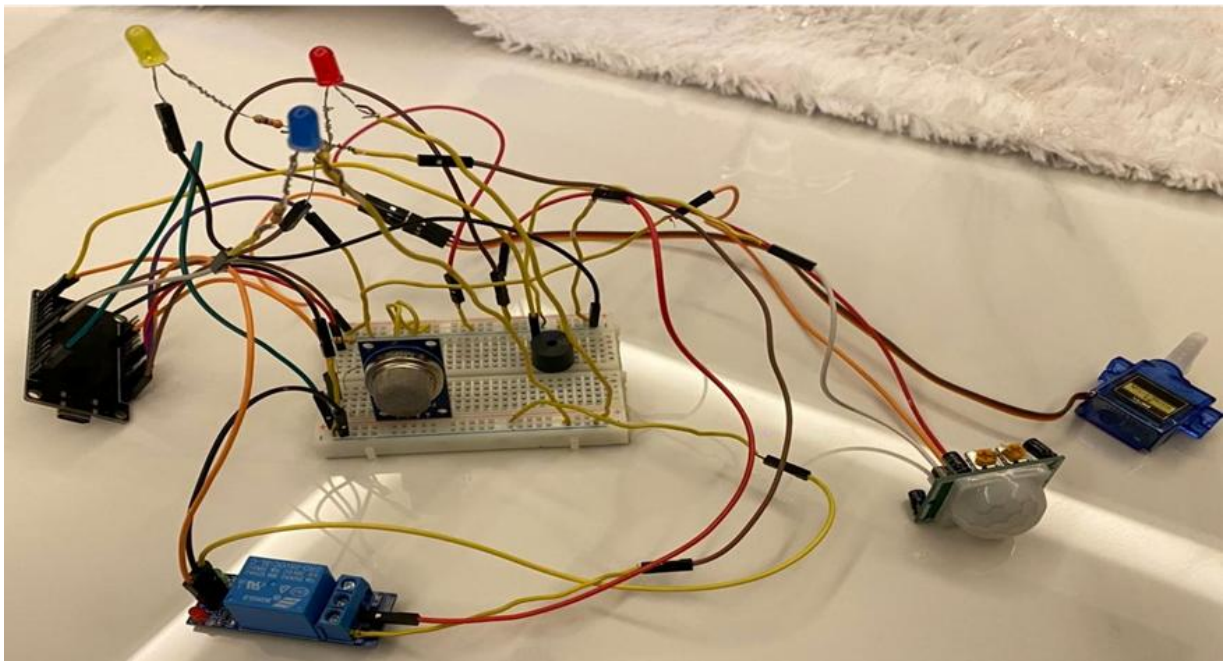


Figure 5.14: Component Integration and Physical Assembly

D. Home Model rough design: The rough design of the home model on paper before actual Building.

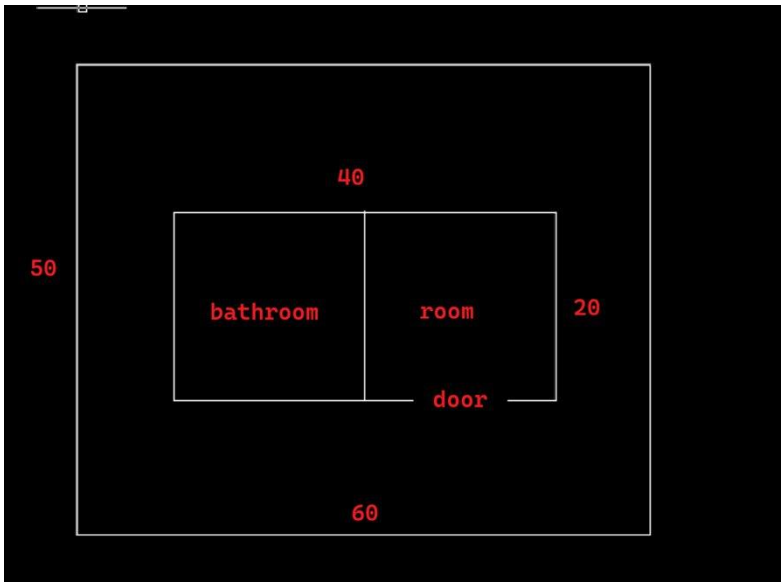


Figure 5.15: Home Model rough design on AutoCAD

E. Home Model Integration: Components installed in a physical model simulating real deployment scenarios

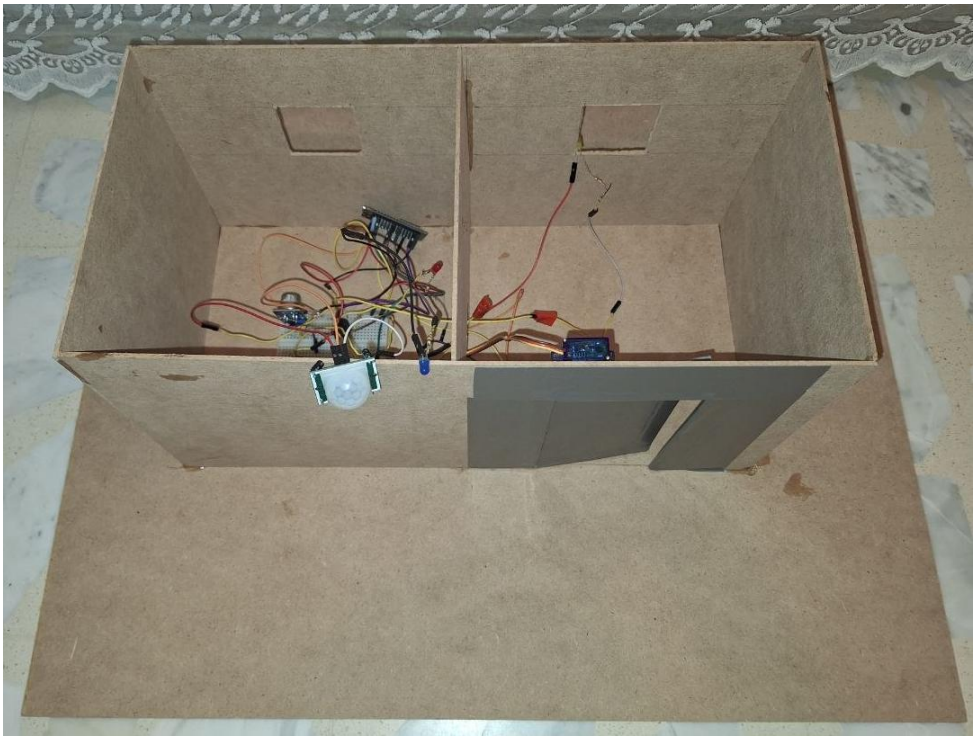


Figure 5.16: The Home Model

The transition from breadboard to permanent installation involved careful positioning of components to mimic their intended real-world roles. The servo motor was mounted to simulate door operation, while sensors were placed to monitor specific areas of the home model, the gas levels and motion detection, creating a demonstration environment that realistically represented the system's practical application.

5.4.2 Circuit Implementation and Testing Process

Circuit development followed an iterative validation approach that ensured reliability at each implementation stage. The process began with individual component verification, progressed through subsystem integration, and culminated in complete system testing within the home model environment.

Implementation Stages:

1. Component-Level Validation: Each sensor and motor tested independently with the ESP32 to insure each was working as intended
2. Subsystem Integration: Sensor cluster and motor validated as coordinated units
3. Full System Assembly: All components connected following documented pin assignments
4. Home Model Deployment: Final installation with optimized cable management and positioning

Critical Implementation Considerations:

- Signal Integrity: Careful routing to minimize noise and interference
- Wire Positioning: Adequate placement of wires for all active components while not causing clutter and disorganization
- Physical Layout: Strategic placement of components in the home model for both functionality and accessibility
- Testing Protocol: Comprehensive validation from individual components to complete workflows

The implementation successfully balanced technical requirements with practical considerations, creating a system that not only functioned correctly but also presented professionally for demonstration purposes.

5.4.3 Firmware Development and System Integration

Firmware development proceeded in parallel with hardware implementation, following a co-development strategy that enabled immediate validation of each new capability. The software architecture centered on a continuous control loop balancing sensor monitoring with command processing while maintaining responsiveness to immediate events.

Core Implementation Features:

Sensor Interface Development:

- MQ-2 Integration: Analog reading calibration through environmental testing
- PIR Configuration: Sensitivity adjustment based on home model characteristics

- Threshold Optimization: Values tuned for realistic home security scenarios
- Presence-Based Activation: Motion sensor monitoring is only active when the system is in ‘away’ mode (triggered via the "Outside Home" command). Upon detection, the ESP32 triggers a sequential lighting pattern across three LEDs as a visual alert, providing clear indication of motion events
- Response Timing: Alert generation latency minimized for safety-critical functions

Motor Control Implementation:

- Servo Positioning: Smooth motion profiles for realistic door simulation
- Alert Coordination: Buzzer and LED patterns synchronized with sensor events
- Command Processing: Efficient parsing of MQTT messages for immediate response
- State Management: Reliable tracking of system status across operation modes

Hardware and Firmware Integration Results:
The hardware and firmware integration successfully demonstrated seamless interaction between all system components. The implementation achieved appropriate response latencies for security applications and validated the practical applicability of the design through realistic testing scenarios within the home model environment. The comprehensive documentation maintained throughout development supports future enhancements and provides clear guidance for system maintenance and troubleshooting.

The individual component implementations now converge in the system integration phase, where we validate complete workflows spanning all three architectural tiers.

5.5 System Integration and Testing

The system integration phase involved methodically connecting and validating the interaction between the three main subsystems: the Flutter mobile application, Python Flask backend server, and ESP32 hardware module. This process focused on establishing reliable communication pathways and verifying that core security workflows functioned as designed.

5.5.1 Connection Establishment and Basic Validation

Step 1: Hardware and firmware activation and connecting them to the MQTT broker
Integration began with sequential component activation to isolate and resolve connectivity issues before attempting complete system operation. The ESP32 hardware was powered on first, establishing its Wi-Fi connection to the home network, connecting to the local MQTT broker and running the firmware on Arduino IDE successfully. Serial monitor output confirmed successful network association and MQTT connection, while the onboard LEDs provided simple visual status indication.

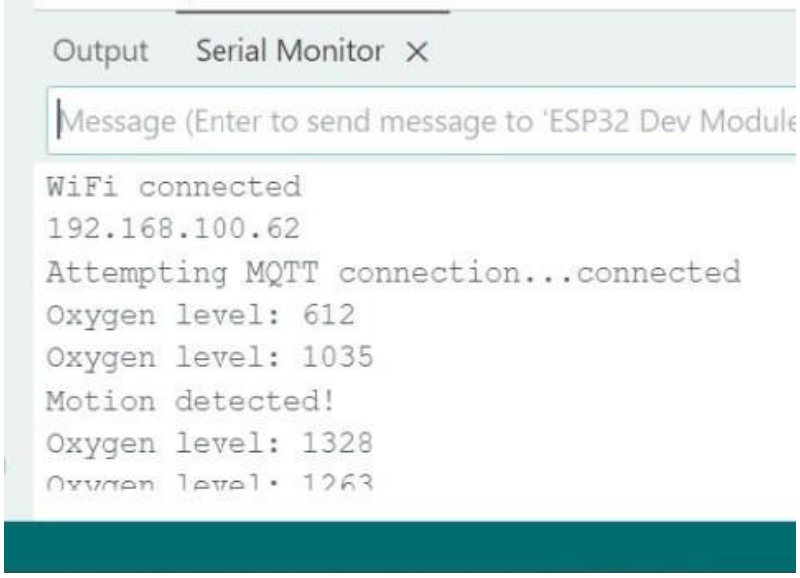


Figure 5.17: Successful connection between MQTT and Arduino IDE

Step 2: Running the flask backend server and establishing its MQTT connection

Once the hardware was communicating, the Flask backend server was started. The server loaded its configuration, initialized the facial recognition libraries, and established its own MQTT client connection to subscribe to hardware alerts using the server's IP address. The server's console output confirmed it was listening on port 5000 for HTTP requests while simultaneously monitoring MQTT topics for sensor events once activated.

```
PS C:\WINDOWS\system32> & "C:\Program Files\mosquitto\mosquitto.exe" -c "C:\Users\96277\Desktop\mosquitto1.conf" -v
>>
1765546056: mosquitto version 2.0.22 starting
1765546056: Config loaded from C:\Users\96277\Desktop\mosquitto1.conf.
1765546056: Opening ipv4 listen socket on port 1884.
1765546056: mosquitto version 2.0.22 running
1765546065: New connection from 192.168.100.15:49951 on port 1884.
1765546065: New client connected from 192.168.100.15:49951 as FlaskServer (p2, c1, k60).
1765546065: No will message specified.
1765546065: Sending CONNACK to FlaskServer (0, 0)
1765546065: New connection from 192.168.100.15:49954 on port 1884.
1765546065: New client connected from 192.168.100.15:49954 as FlaskOxygenSubscriber (p2, c1, k60).
1765546065: No will message specified.
1765546065: Sending CONNACK to FlaskOxygenSubscriber (0, 0)
1765546065: Received SUBSCRIBE from FlaskOxygenSubscriber
1765546065:     home/oxygen_alert (QoS 0)
1765546065: FlaskOxygenSubscriber 0 home/oxygen_alert
1765546065: Sending SUBACK to FlaskOxygenSubscriber
1765546065: Received SUBSCRIBE from FlaskOxygenSubscriber
1765546065:     home/motion_alert (QoS 0)
1765546065: FlaskOxygenSubscriber 0 home/motion_alert
1765546065: Sending SUBACK to FlaskOxygenSubscriber
```

Figure 5.18: Successful Connection Between MQTT and Flask

Step 3: Activating the Flutter app

Finally, the mobile application was launched on a physical Android device using flutter build apk --release. The app initialized its Firebase components, requested necessary camera and storage permissions, and attempted to connect to the Flask server's IP address. Successful connection was indicated by the home screen loading with functional buttons for registration and authentication.

```
Running Gradle task 'assembleDebug'...                             411.7s
✓ Built build\app\outputs\flutter-apk\app-debug.apk
Installing build\app\outputs\flutter-apk\app-debug.apk...           56.0s
D/FlutterJNI( 9554): Beginning load of flutter...
D/FlutterJNI( 9554): flutter (null) was loaded normally!
I/flutter ( 9554): [IMPORTANT:flutter/shell/platform/android/android_context_
ering backend (Vulkan).
I/Choreographer( 9554): Skipped 46 frames!  The application may be doing too
```

Figure 5.19: Successful flutter app launch

```
C:\Users\96277\Desktop\connection app\backend>python server4.py
Notification sent: 200 {
  "name": "projects/smarthomeprojectabj/messages/7572141554479624485"
}

Connected to MQTT for Oxygen alerts, code: 0
 * Serving Flask app 'server4'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.100.15:5000
Press CTRL+C to quit
192.168.100.11 - - [12/Dec/2025 16:32:55] "GET / HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:33:18] "POST /recognize HTTP/1.1" 200 -
```

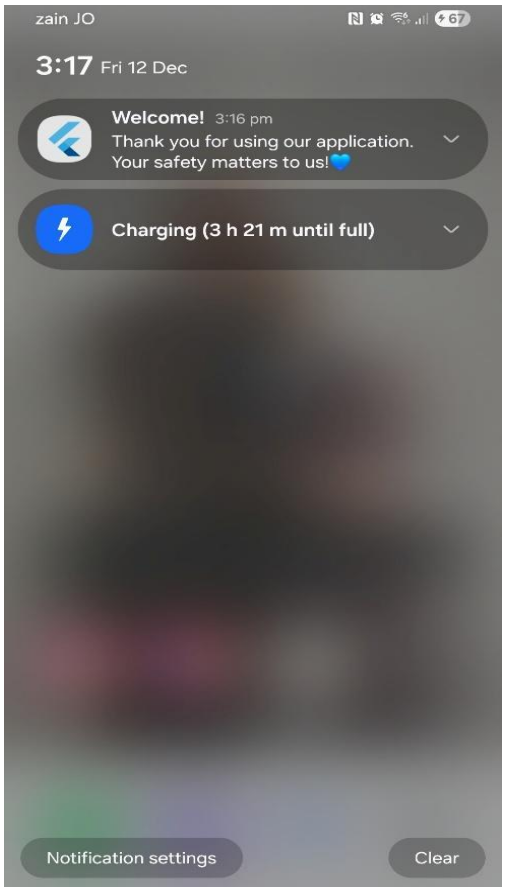
Figure 5.20: Successful Connection Between Server and App

This step-by-step approach allowed each component to be validated independently before attempting complex interactions, making troubleshooting more manageable when issues arose.

5.5.2 Communication Protocol Validation.

With all components active, the integration focused on verifying the three communication layers that enable system functionality.

Multi-user connection confirmation:
Upon Activating the server for the first time, the FCM pushes a ‘thank you’ notification to all users that have the full released app, confirming not only the firebase is active and working, but also that all app users are connected and will be able to receive important notifications and use the server. From that point forward, any notification from the hardware regarding gas alerts or motion alerts will be received effectively by all users.



52 Figure 5.21: Thank you FCM notification for all users

Mobile-to-Server HTTP Communication:

The RESTful API design was validated through basic endpoint testing. Registration attempts confirmed that multipart form data containing both credentials and facial images were correctly received and processed by the Flask server. Authentication sequences verified that recognition requests and password verification operated as expected, with appropriate JSON responses returned to the mobile application. Control commands for door operation and system functions demonstrated that simple POST requests could trigger backend actions. Every POST and GET appearing in the server's command and MQTT line reaffirms the success of the connection.

```
C:\Users\96277\Desktop\connection app\backend>python server4.py
Notification sent: 200 {
  "name": "projects/smarthomeprojectabj/messages/7521191394250977500"
}

Connected to MQTT for Oxygen alerts, code: 0
 * Serving Flask app 'server4'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.100.15:5000
Press CTRL+C to quit
192.168.100.11 - - [12/Dec/2025 16:38:43] "GET / HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:39:05] "POST /recognize HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:39:09] "POST /verify_password HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:39:10] "GET /dataset HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:39:12] "POST /open_door HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:39:15] "POST /close_door HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:40:33] "POST /motion_inside HTTP/1.1" 200 -
192.168.100.11 - - [12/Dec/2025 16:40:38] "POST /motion_outside HTTP/1.1" 200 -
```

Figure 5.22: Hardware control massages, from app to MQTT to the hardware.

Server-to-Hardware MQTT Communication:

The MQTT publish-subscribe model was tested by sending command strings from the Flask server to the ESP32 and verifying hardware responses. Messages published to the home/door topic with "open" and "close" payloads triggered corresponding servo movements. Simultaneously, the server's subscription to home/oxygen_alert, home/motion_alert, and door command topics were validated by having the ESP32 publish test messages that appeared in the Flask console output.

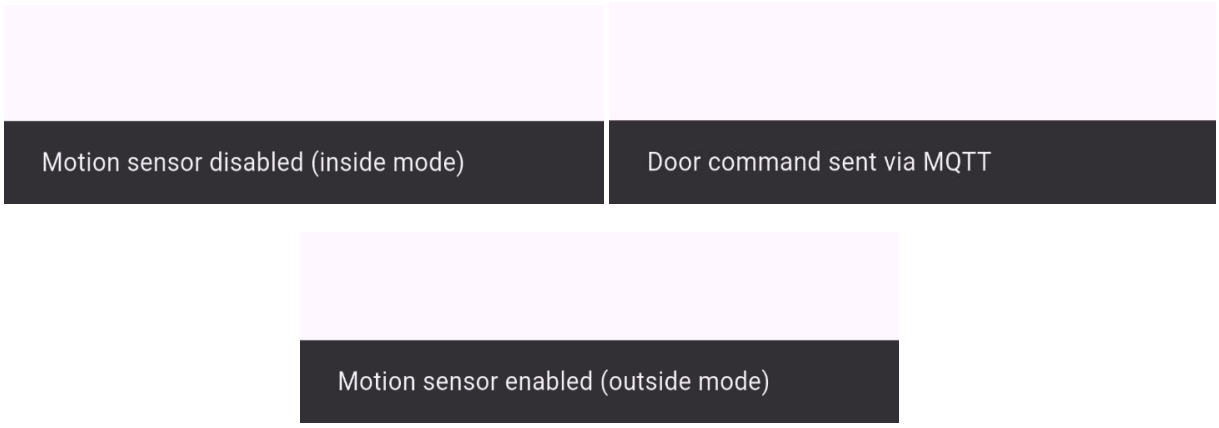
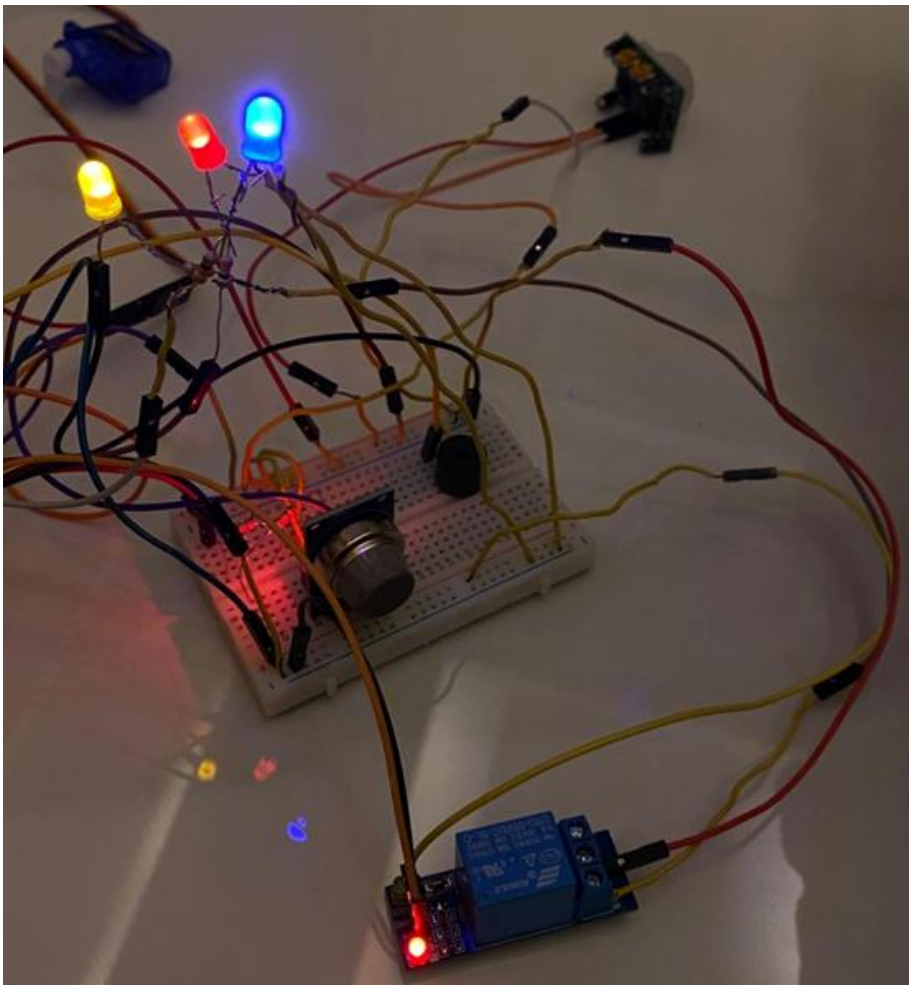


Figure 5.23: Snack bar showing the hardware response success

Local Hardware Communication:

At the device level, the ESP32's interaction with connected sensors and actuator was verified. Analog readings from the MQ-2 sensor showed expected value ranges, digital inputs from the PIR sensor changed state with motion detection, and PWM signals successfully positioned the servo motor. These local operations confirmed that the hardware could both monitor its environment and execute physical actions.



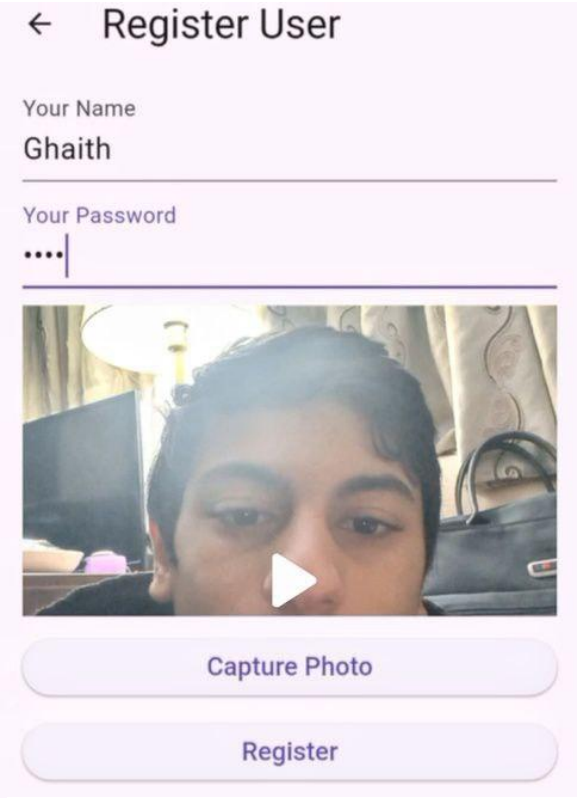
55 *Figure 5.24: Motion sensor working as intended*

5.5.3 Core Workflow Validation

Integration testing focused on complete user scenarios that spanned multiple components and communication methods.

User Registration and Enrollment:
The complete registration process was tested: capturing a facial image through the mobile app camera, transmitting it with credentials to the Flask server, processing and storing the facial encoding, and returning a success confirmation. This validated the end-to-end data flow from image capture through server processing to persistent storage.

Authentication and Access Control:



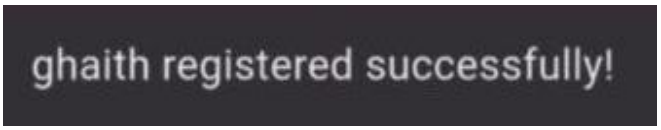


Figure 5.25: Registration page: Password and Image Insertion

The two-factor authentication sequence was verified: facial recognition through image capture and server-side comparison followed by password entry and verification. Successful authentication granted access to the security control dashboard, while failed attempts provided appropriate error messages. This testing confirmed the security model operated as intended.

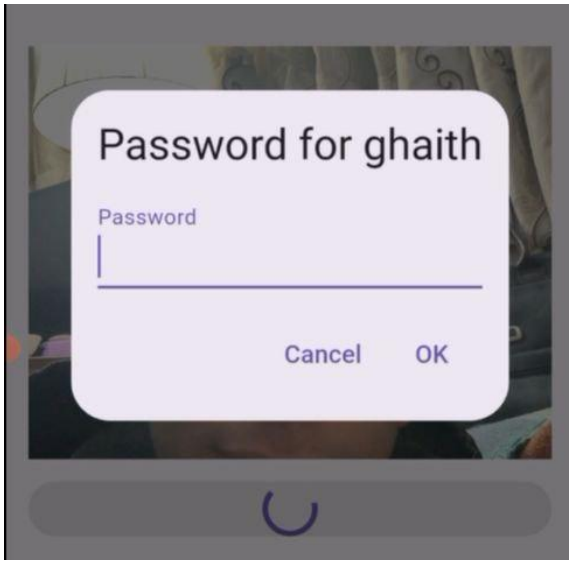


Figure 5.26: Recognition successful, enter the password

Security Monitoring and Response:
The system's ability to detect and respond to potential hazards was validated. Simulated oxygen level increases triggered the MQ-2 sensor threshold, causing the ESP32 to publish an alert that the Flask server received and the buzzer to activate, providing a physical alert heard throughout the home. Motion detection events were configured to operate only when the user is away: pressing the "Outside Home" button in the app activates motion monitoring, while pressing "Inside Home" deactivates it. When active and motion is detected, the system triggers three LED indicators to light up sequentially, providing a clear visual alert. These tests confirmed that the monitoring infrastructure functioned correctly and responded appropriately to user presence settings.

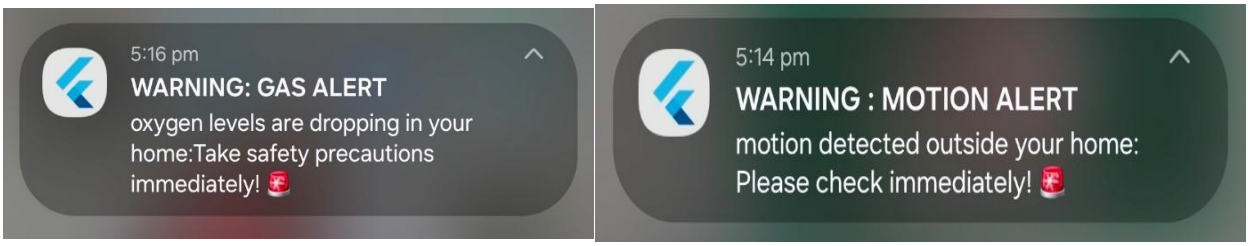


Figure 5.27: Gas and Motion Alert Group notification

Remote Control Operation:
Hardware control from the mobile interface was tested through door open/close commands. The complete path—mobile app button press, HTTP request to Flask, MQTT message to ESP32, servo actuation—was verified to operate reliably with appropriate timing for user feedback.

5.5.4 Firebase Cloud Messaging Configuration

Firebase Cloud Messaging infrastructure has been configured as an extensible notification layer.

The Flask backend contains complete Firebase service account authentication with OAuth2 token management properly implemented. The `send_fcm_notification()` function structure has been tested to successfully communicate with Firebase servers when manually invoked. Topic-based subscription management is conceptually established for categorized alert delivery, though the current implementation focuses on local network communication.

This FCM configuration represents a scalable foundation that could support cloud-based notifications reaching users beyond their home network, providing a clear migration path from the current local-only alerting to broader notification scenarios in future development phases.

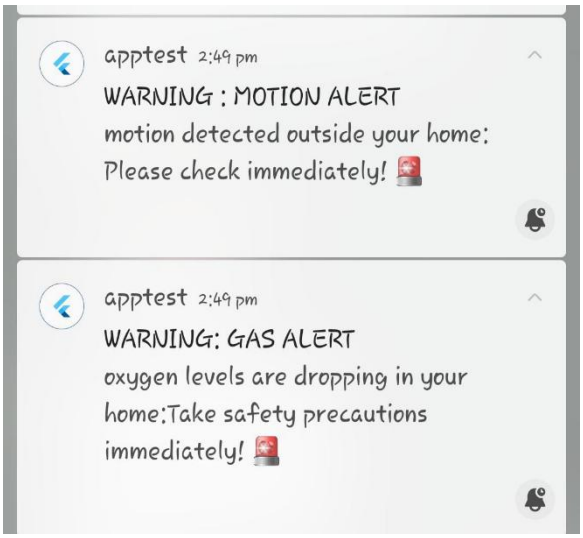


Figure 5.28: The same notifications from a different user

5.5.5 Integration Outcomes and System Validation

The integration process successfully demonstrated that all system components can communicate effectively to provide core home security functionality. The mobile application provides intuitive access to security features, the Flask backend reliably processes authentication and coordinates communications, and the ESP32 hardware effectively monitors the environment and executes physical controls.

Key Integration Achievements:

- Reliable Communication: All three communication layers (HTTP, MQTT, local GPIO) function as designed
- Complete Workflows: End-to-end user scenarios operate successfully from start to finish
- Appropriate Responsiveness: System reactions occur within timeframes suitable for security applications
- Clear User Feedback: Success and error states are communicated effectively through the interface
- Extensible Architecture: The system design accommodates both current functionality and future enhancements

The integrated prototype validates the fundamental concept of an affordable, accessible smart home security system that combines facial recognition authentication, environmental monitoring, and remote-control capabilities into a cohesive platform suitable for typical household deployment.

With all components operational and integrated, we now present the results that showcase the system's capabilities and validate its design decisions.

5.6 Results

5.6.1 Interoperability Testing:

- Component Compatibility: All hardware and software components interoperate without conflicts
- Protocol Compliance: Full adherence to HTTP, MQTT, and serial communication standards
- Data Consistency: Information remains consistent across all system layers
- Error Propagation: Errors are properly handled and communicated through the system

5.6.2 Performance Evaluation

Percentage Tested Accuracy with our Dataset:

- Detection accuracy: 93.75%
- Recognition accuracy: 100%

```
75  print(f"Faces detected: {result['faces_detected']}")
76  print(f"Faces recognized correctly: {result['faces_recognized_correctly']}")
77  print(f"Detection accuracy: {result['detection_accuracy']}%")
78  print(f"Recognition accuracy: {result['recognition_accuracy']}%")

PROBLEMS 8   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\96277\Desktop\connection app> & C:/Users/96277/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/96277/Desktop/
test.py"
PS C:\Users\96277\Desktop\connection app> & C:/Users/96277/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/96277/Desktop/
test.py"
{'total_images': 16, 'faces_detected': 15, 'faces_recognized_correctly': 15, 'detection_accuracy': 93.75, 'recognition_accuracy': 100.
Total images: 16
Faces detected: 15
Faces recognized correctly: 15
Detection accuracy: 93.75%
Recognition accuracy: 100.0%
```

Figure 5.29: Recognition and detection accuracy test across our dataset

Note: Recognition Model has been pretrained on huge datasets and has the tested accuracy of 99.38%, the accuracy above was a result on testing the model on our dataset, that was collected manually.

Qualitative Performance Assessment:

1. Usability:
    - Clear feedback for all user actions
    - Comprehensive error messages
    - Accessible design for users with varying technical expertise

2. Security:
    - Multi-factor authentication provides strong access control
    - Encrypted communications protect sensitive data
    - Audit logging for all security-relevant events

5.6.3 Discussion

Achievements and Successes:

1. Technical Integration: Successfully integrated diverse technologies (Flutter, Flask, ESP32, MQTT, Firebase) into a cohesive system
2. Reliability: Demonstrated robust operation under various environmental conditions
3. Scalability: Architecture supports future expansion with additional sensors and features

Comparative Analysis:
Compared to existing solutions documented in Chapter 2, our system demonstrates:

- Superior Integration: Combines multiple safety features in a single platform
- Better Affordability: Lower cost than commercial security systems with similar capabilities
- Enhanced Usability: More intuitive interface than most DIY security solutions
- Greater Flexibility: Modular design allows customization for different home environments

Validation of Design Decisions:

1. Hybrid Architecture: The decision to split processing between mobile, server, and hardware was validated
2. MQTT Protocol: Provided reliable, low-latency communication as evidenced by testing metrics
3. Modular Design: Facilitated testing, debugging, and future enhancements

This chapter has systematically documented the development and validation of the integrated home security system. The implementation successfully demonstrates: (1) an intuitive mobile interface for user interaction, (2) a robust facial recognition backend for secure authentication, and (3) responsive hardware for environmental monitoring and physical control. Through phased testing and integration, the system has proven its capability to address real-world home safety concerns while maintaining scalability for future expansion. These achievements establish a working prototype that both validates the project's design decisions and provides measurable results for evaluation.

Chapter 6: Conclusions and Future Work

This chapter synthesizes the project's achievements, discusses the implications of the implemented system, and outlines potential directions for future development and enhancement. It presents both the concrete outcomes of the current work and a roadmap for its evolution based on current technological trends and identified limitations.

6.1 Conclusions

The General Safety and Security Home System project has successfully demonstrated the feasibility and effectiveness of an integrated smart home security solution. Through meticulous design, implementation, and testing across multiple technological domains, the project has achieved its primary objectives of enhancing home safety through a practical, accessible technological approach.

System Architecture and Integration Success
The project's most significant achievement lies in its successful integration of three distinct technological layers into a cohesive operational unit. The mobile application built with Flutter provides an intuitive user interface, creating a seamless user experience from authentication to control **[27].** The Flask backend server effectively processes facial recognition requests while coordinating communication between components through both HTTP/REST APIs and MQTT protocols. The ESP32 hardware implementation demonstrates practical IoT deployment with reliable sensor monitoring and motor control, validating the use of affordable, accessible components for home security applications **[36 - 41].**

Technical Innovation and Practical Implementation
Beyond mere component integration, the project introduces several innovative approaches to home security. The two-factor authentication system combining facial recognition with password verification addresses both convenience and security concerns effectively **[15].** The hybrid communication model using MQTT for local hardware control alongside HTTP for mobile-server communication provides an optimal balance of responsiveness and reliability **[44, 45].** Most importantly, the system validates that comprehensive security features can be implemented using components totaling approximately 25 Jordanian Dinars, making advanced home security accessible to average households **[Table 3.1].**

Educational and Development Value
From a learning perspective, the project provided comprehensive experience across multiple modern technology domains. The development process encompassed mobile application development with Flutter, server-side programming with Python and Flask, computer vision implementation with OpenCV and face_recognition libraries, IoT hardware integration with ESP32, and network communication protocols including HTTP, MQTT, and serial communication. This multidisciplinary approach reflects real-world engineering challenges and solutions.

Performance Validation

Testing confirmed that all core system requirements have been successfully met. The facial recognition system achieves practical accuracy suitable for home access control, the environmental monitoring reliably detects safety hazards, and the hardware control system responds within appropriate timeframes for security applications. Perhaps most importantly, the system demonstrates reliable interoperability between components—a critical factor often challenging in IoT implementations **[Chapter 5.5].**

6.2 Future Work

The current implementation provides a solid foundation for numerous enhancements and expansions, organized into three development horizons based on complexity, resource requirements, and technological readiness.

6.2.1 Short-term Improvements (0-6 months)

Enhanced Recognition Algorithms

While the current facial recognition system using HOG features and 128-dimensional encodings provide adequate accuracy **[15],** several improvements could significantly enhance reliability. Implementing deep learning-based approaches, particularly convolutional neural networks (CNNs) for feature extraction, could improve accuracy in challenging conditions such as low lighting or partial occlusions **[related to 2.4, 2.5].** Additionally, incorporating liveness detection techniques would prevent spoofing attempts using photographs or videos, a known vulnerability in facial recognition systems **[4].**

Extended Sensor Capabilities

The current environmental monitoring focuses on combustible gases and motion detection. Expanding this suite to include carbon monoxide (CO) sensors would address another critical home safety hazard, particularly relevant with gas heating systems **[related to 2.2]**. Water leak detection sensors placed near water heaters, washing machines, and under sinks could prevent costly water damage. Window and door contact sensors would provide more comprehensive perimeter security than motion detection alone.

Data Management Evolution - Local Database Migration

The current file-based storage system using pickle files and directory structures could be enhanced by migrating to a local SQLite database. This would provide several immediate benefits: more efficient querying of user data and event logs, better data integrity through transactional operations, and standardized data access patterns. SQLite integration would maintain the system's local operation while offering structured data management suitable for modest user counts and event volumes.

Cloud Integration Foundations

While maintaining local operation as primary, implementing optional cloud synchronization would enable backup and remote access features. A lightweight synchronization service could periodically upload anonymized event logs and user metadata (excluding facial images for privacy) to a cloud database, providing data redundancy and enabling multi-device access to security history.

User Experience Refinements
The current interface could be enhanced with voice control integration through platforms like Google Assistant or Amazon Alexa, providing hands-free operation particularly useful in emergency situations. Family sharing features would allow multiple household members to manage security settings with appropriate permission levels. These improvements would make the system more accessible to users with varying technical expertise and physical abilities.

6.2.2 Medium-term Developments (6-18 months)

Advanced Analytics and Intelligence
Moving beyond simple threshold-based alerts, machine learning algorithms could analyze sensor data patterns to identify normal versus abnormal activity, reducing false positives and providing more intelligent alerts **[3].** Predictive maintenance algorithms could monitor hardware performance and alert users to potential failures before they occur. Behavioral analysis could learn household routines to better distinguish between normal activity and potential security threats.

Extended Connectivity and Interoperability
5G connectivity would provide faster, more reliable remote access with lower latency, particularly valuable for real-time video monitoring if cameras are added. Mesh networking capabilities would extend coverage throughout larger properties without signal degradation. Perhaps most importantly, interoperability with existing smart home ecosystems (Google Home, Apple HomeKit, Samsung SmartThings) would allow the security system to work seamlessly with other home automation devices **[6, 7].**

Cloud Data Management and Analytics
Implementing a comprehensive cloud data platform would enable advanced features beyond local capabilities. Cloud storage of encrypted facial encodings (not raw images) would allow recognition across multiple properties or devices. Time-series databases could store sensor data for long-term trend analysis and predictive insights. Data warehousing could aggregate anonymized data from multiple installations to identify broader security patterns and improve system intelligence.

Commercial and Accessibility Features
For broader deployment, professional monitoring service integration would provide 24/7 oversight with emergency response coordination. Insurance company API integration could potentially reduce homeowners' insurance premiums when security systems are installed—a growing trend in the insurance industry. Enhanced accessibility features, including multi-language support and interfaces designed for users with visual or mobility impairments, would ensure the system serves diverse populations effectively.

6.2.3 Long-term Vision (18+ months)

Autonomous and Proactive Security Systems
Looking further ahead, drone integration could provide external property monitoring and automated perimeter patrols, particularly valuable for larger properties. Robotic units could investigate alerts and provide visual verification before human intervention is required. These autonomous systems represent the future of integrated security where technology not only alerts users to issues but can investigate and respond appropriately.

Distributed and Edge Data Architecture
Future systems could implement edge computing for data processing, where initial analysis occurs on local devices with only significant events or aggregated insights transmitted to central systems. This distributed approach would improve privacy (less data transmission), reduce bandwidth requirements, and increase system resilience during network outages. Blockchain technology could provide immutable audit logs for security events, particularly valuable for insurance or legal purposes.

Community and Networked Security
Future developments could transform individual home security systems into networked community safety platforms. Neighborhood security networks could share anonymized threat data, creating community-wide awareness of security concerns. Integration with public safety systems could provide direct emergency service notification with verified information about the nature and location of incidents.

Sustainability and Environmental Integration
As environmental concerns grow, security systems could incorporate sustainability features. Solar power integration with battery storage would provide operation during power outages—a critical feature for security systems. Energy harvesting from environmental sources could further reduce dependence on external power. Biodegradable or recyclable component design would address electronic waste concerns associated with IoT device proliferation.

Research Directions in Security Technology
Longer-term research could explore cutting-edge security technologies. Privacy-preserving facial recognition using techniques like federated learning would allow recognition without centralized storage of biometric data. Quantum-resistant encryption would future-proof communications against emerging computational threats. Neuromorphic computing could enable more sophisticated on-device processing with dramatically reduced power consumption compared to traditional approaches.

6.3 Final Remarks

The General Safety and Security Home System project represents more than a technical implementation—it demonstrates a practical approach to making advanced security technology accessible and manageable for typical households. Beyond the specific features implemented, the

project offers valuable insights into the integration challenges and design considerations for IoT-based home automation systems.

Practical Engineering Insights
The development process revealed several important considerations for similar projects. The balance between local processing (for responsiveness) and cloud capabilities (for scalability) proved crucial, as did the careful selection of communication protocols based on specific use cases. The decision to use MQTT for hardware communication while employing HTTP for mobile-server interactions created an effective hybrid architecture that could be adapted to various home networking environments.

Design Philosophy Validation
The project validates an important design philosophy: that sophisticated technology need not be complex to use. By maintaining focus on intuitive interfaces while implementing advanced backend capabilities, the system demonstrates how technical sophistication and user-friendliness can coexist. This approach has broader implications for consumer technology development, particularly in domains where users may not have technical expertise but still benefit from advanced features.

Educational and Developmental Value
From an educational perspective, the project successfully integrated knowledge from multiple domains -mobile development, computer vision, embedded systems, and network protocols- into a cohesive whole. This multidisciplinary integration mirrors real-world engineering challenges and provides a model for comprehensive project-based learning in technology education.

Forward-Looking Perspective
As smart home technologies continue to evolve, projects like this contribute to an important conversation about the role of technology in daily life. The balance between capability and usability will become increasingly important as more aspects of home management become automated and connected.

Closing Reflection
The true measure of this project's success lies not only in its technical implementation but in its potential to make genuine improvements to home safety and user peace of mind. By addressing specific, well-defined safety concerns with practical technological solutions, the project demonstrates how focused engineering can create meaningful benefits for everyday life. As technology continues to advance, maintaining this connection between technical capability and human need remains essential for creating systems that truly enhance quality of life.

The project stands as evidence that with careful design, thoughtful implementation, and user-centered development, advanced security technology can be made both accessible and effective for average households, providing both immediate safety benefits and a pathway toward more intelligent, integrated, and responsive home environments in the future.

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Appendix

Appendix A: Data Definition Tables

Table A.1: App tool definition

Category	Formal Definition & Key Characteristics	Primary Role in the Project	
Flutter	Framework	An open-source UI software development toolkit created by Google for building natively compiled applications across mobile, web, and desktop from a single codebase.	Core framework for developing the cross-platform mobile application.
Dart	Programming Language	A client-optimized language for fast apps on multiple platforms; primary language for Flutter, featuring sound null safety.	Used to write application logic, UI construction, business rules, and network communication.
Visual Studio Code	Development Tool	A lightweight, extensible source code editor by Microsoft, enhanced via extensions.	Primary IDE for Flutter development, debugging, and project management.
Flutter SDK	Software Development Kit	Collection of tools, libraries, and APIs for Flutter development, including engine, widgets, and CLI tools.	Essential for compiling, building, and running the Flutter app.
Widget	UI Component	Fundamental building block of Flutter UI; everything from layout to style is a widget.	Used to construct the entire mobile UI in a composable manner.
Hot Reload	Development Feature	Flutter feature that injects updated code into a running app, updating	Accelerated development and UI iteration.

		the UI almost instantly without restart.	
flutter doctor	Command-Line Tool	Flutter CLI command that checks the development environment and reports on the status of dependencies.	Validated correct installation of prerequisites (Android SDK, etc.).
pubspec.yaml	Configuration File	Central file for managing Flutter project dependencies, assets, and metadata.	Declared external packages and project settings.
Flask	Backend Framework	Lightweight Python web framework for building RESTful APIs and handling HTTP requests.	Backend server for user registration, authentication, facial recognition, and hardware communication.
USB Debugging	Android Feature	Developer mode enabling communication between Android device and computer for advanced operations.	Enabled direct deployment and testing of the Flutter app on physical devices.
Face Recognition Module	Backend Component	Uses Python's face_recognition and OpenCV libraries to extract and compare facial encodings.	Handles facial registration and user recognition.
Dataset Handling System	Backend Component	Manages facial images and encodings stored in structured directories and serialized .pkl files.	Stores and retrieves registered user data for recognition tasks.
OpenCV (cv2)	Computer Vision Library	Open-source library for real-time computer vision and image processing.	Used for image manipulation, face detection, cropping, resizing, and preprocessing.
face_recognition	Python Library	Python wrapper for dlib's facial recognition capabilities; uses HOG + SVM for detection and ResNet-34 for encoding.	Core library for face detection and generating 128-dimensional facial embeddings.

NumPy	Numerical Library	Library for numerical computing in Python, supporting multi-dimensional arrays and mathematical operations.	Used for matrix operations and handling image data in facial recognition.
Paho MQTT	Communication Library	Python client for the MQTT protocol, enabling publish/subscribe messaging.	Facilitates real-time communication between Flask backend and ESP32 hardware.
Mosquitto	MQTT Broker	Open-source message broker that implements the MQTT protocol.	Local broker used for handling MQTT messages between Flask and ESP32.
ESP32-WROOM	Microcontroller	Dual-core 240 MHz microcontroller with Wi-Fi and Bluetooth, 520 KB RAM, 4 MB flash, 3.3V operation.	Central hardware unit interfacing with sensors, actuators, and communicating with the backend via MQTT.
MQ-2 Sensor	Gas Sensor	Metal oxide semiconductor sensor for detecting flammable gases and smoke (LPG, propane, methane, etc.). Analog and digital outputs.	Monitors indoor air for dangerous gas levels; triggers alerts.
SG90 Servo Motor	Actuator	Micro servo with 180° rotation, 5V operation, controlled via PWM signal.	Simulates door lock/unlock mechanism based on authentication results.
PIR Motion Sensor	Motion Sensor	Passive infrared sensor for detecting human motion; digital output, 5V operation, adjustable range and delay.	Detects motion when system is in “away” mode; triggers LED alerts.
Active Buzzer	Alert Device	5V buzzer that produces sound when activated with a HIGH signal.	Audible alert for gas detection or security breaches.

Relay Module	Switching Device	Electromechanical switch controlled by low-voltage signal; used to control higher-voltage devices.	Controls AC/DC devices (e.g., lights) for occupancy simulation.
LED	Indicator	Light-emitting diode used for visual signaling; operates at 2–3.3V with current limiting resistor.	Visual alerts for motion detection, system status, or warnings.
Firebase Cloud Messaging (FCM)	Cloud Service	Google's cross-platform messaging solution for sending notifications to mobile apps.	Sends push notifications to the Flutter app for alerts (oxygen, motion).
Google OAuth2	Authentication Library	Google's authentication library for server-to-server communication via service accounts.	Authenticates backend with Firebase services for sending notifications.
HTTP Package (Dart)	Networking Library	Dart package for making HTTP requests.	Used in Flutter app to communicate with Flask backend (registration, login, control commands).
Image Picker (Flutter)	Flutter Plugin	Plugin for picking images from device gallery or camera.	Enables users to upload facial images for registration and recognition.
Image Cropper (Flutter)	Flutter Plugin	Plugin for cropping images before upload.	Improves face recognition accuracy by allowing users to crop images to focus on the face.
Firebase Core (Flutter)	Flutter Plugin	Required for initializing Firebase services in Flutter.	Initializes Firebase for notifications and analytics.

Firebase Messaging (Flutter)	Flutter Plugin	Enables receiving push notifications via FCM in Flutter apps.	Handles receipt and display of security alerts in the app.
Flutter Local Notifications	Flutter Plugin	Displays local notifications within the app.	Shows in-app alerts for system events when the app is in the foreground.

Appendix B: Hardware Description

Table B.1: ESP32 DevKIT (WROOM) pin color coordination and characteristics

Color	**Function**	**Examples / Notes**
Purple	GPIO Pins (Digital / ADC)	GPIO0, GPIO2, GPIO4, GPIO12–15, GPIO25–27, GPIO32–39
Orange	RTC GPIO / Touch Pins / Input Only	Touch0–Touch9, Input Only Pins: GPIO34, 35, 36, 39, RTC Pins: GPIO0, 3, 4, 5, 9, 10, 12–15, 16, 17
Brown	DAC Pins	GPIO25 (DAC1), GPIO26 (DAC2)
Gray	Power Pins	GND, VIN, 3.3V
Green	Touch Pins	Touch0, Touch2, Touch4, Touch5, Touch6, Touch7, Touch8, Touch9
Light Yellow	SPI Pins (MOSI, MISO, CLK, CS)	HSPI / VSPI Pins, e.g., GPIO13–15, GPIO18–23, GPIO16, GPIO14 (depends on HSPI / VSPI mapping)
Light Blue / Cyan	SPI Buses / SPI Function	VSPI / HSPI lines, e.g., MOSI, MISO, SCK, CS, SDA, SCL
Dark Yellow / Orange-yellow	Special / Boot / Serial	GPIO0, 2, 15 (Boot), GPIO1 (TX0), GPIO3 (RX0)

Table B.2: ESP32 DevKIT (WROOM) pin numerical labels and in/out characteristics

Pin	Pin Type	Analog	Digital	Input	Output	Notes / Usage
0	GPIO0	Yes (ADC2)	Yes	Yes	Yes	Boot mode during startup, avoid using for IO at boot
1	GPIO1	No	Yes	Yes	Yes	TX0, used for Serial, avoid using for basic IO
2	GPIO2	Yes (ADC2)	Yes	Yes	Yes	Often used for LED or I2C SDA
3	GPIO3	No	Yes	Yes	Yes	RX0, used for Serial
4	GPIO4	Yes (ADC2)	Yes	Yes	Yes	Good for sensors like MQ2

5	GPIO5	Yes (ADC2)	Yes	Yes	Yes	Can be used for general purposes
12	GPIO12	Yes (ADC2)	Yes	Yes	Yes	Caution: boot strapping, avoid high load
13	GPIO13	Yes (ADC2)	Yes	Yes	Yes	LED, SPI SCK
14	GPIO14	Yes (ADC2)	Yes	Yes	Yes	Good for Buzzer or Relay control
15	GPIO15	Yes (ADC2)	Yes	Yes	Yes	Caution: boot strapping
16	GPIO16	No	Yes	Yes	Yes	Regular Digital IO
17	GPIO17	No	Yes	Yes	Yes	Regular Digital IO
18	GPIO18	Yes (ADC2)	Yes	Yes	Yes	SPI SCK, can be used as regular IO
19	GPIO19	Yes (ADC2)	Yes	Yes	Yes	SPI MISO or regular IO
21	GPIO21	Yes (ADC2)	Yes	Yes	Yes	I2C SDA
22	GPIO22	Yes (ADC2)	Yes	Yes	Yes	I2C SCL
23	GPIO23	Yes (ADC2)	Yes	Yes	Yes	SPI MOSI or regular IO
25	GPIO25	Yes (ADC1)	Yes	Yes	Yes	DAC1, can output analog signal
26	GPIO26	Yes (ADC1)	Yes	Yes	Yes	DAC2, great for Relay control via transistor
27	GPIO27	Yes (ADC1)	Yes	Yes	Yes	Excellent Analog input
32	GPIO32	Yes (ADC1)	Yes	Yes	Yes	Analog input only (Input only)
33	GPIO33	Yes (ADC1)	Yes	Yes	Yes	Analog input only (Input only)
34	GPIO34	Yes (ADC1)	No	Yes	No	Input only, cannot be used as Output
35	GPIO35	Yes (ADC1)	No	Yes	No	Input only, cannot be used as Output
36	GPIO36	Yes (ADC1)	No	Yes	No	VP, Input only
39	GPIO39	Yes (ADC1)	No	Yes	No	VN, Input only